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Casne CloudStream User Guide

Version 2.0

Introduction

This user guide contains instructions for operating the **CloudStream System Control** application, the web-based interface for controlling **Casne CloudStream**.

CloudStream is a modular data pipeline for the Industrial Internet of Things (IIoT). Our customers use CloudStream's custom data connectors to bridge data between their sensors, industrial data systems, and cloud platforms.

When used as a Software as a Service (SaaS) application, our connectors enabled niche applications that no other turnkey solutions offer – such as connecting AVEVA PI historians directly to ESRI ArcGIS systems, or data concentration between AVEVA PI historians and AVEVA CONNECT Data Services. Our customers find that these applications significantly reduce the complexity and cost of their IIoT infrastructure.

Table of Contents

1. CloudStream Roles	5
2. CloudStream System Control	6
2.1. Installation	6
2.2. System Control Access.....	7
2.3. Node Management.....	8
2.4. Node Configuration (“Config”)	10
2.5. User Management (“Users”)	11
2.6. CloudStream System Control (“System”)	13
2.6.1. CloudStream Control.....	13
2.6.2. Data Injector Subservice Management (Edge Deployments)	14
2.6.3. Data Injector Subservice Management (Cloud Deployments).....	16
2.7. Logging (“Logs”).....	19
3. CloudStream Source Creation.....	20
3.1. AVEVA CONNECT Data Services.....	21
3.2. CSV	25
3.3. Data Generator.....	26
3.4. MQTT	32
3.5. AVEVA PI Web API (Attributes & Elements)	35
3.6. AVEVA PI Web API (Tags).....	41
3.7. AVEVA PI Web API (Event Frames)	45
4. CloudStream Destination Creation	52
4.1. AVEVA CONNECT Data Services.....	53
4.2. Azure Event Hub	59
4.3. ESRI GeoEvent Server	60
4.4. MQTT	61
4.5. AVEVA PI Web API (Tags).....	63
4.6. pSQL.....	67
4.7. Vantiq.....	69
5. Data Injectors.....	70

5.1	Configuring Data Injectors.....	71
5.2	Editing Data Injectors	76
6.	CloudStream Service Creation	78
6.1	Service Monitoring	81
6.2	Service Control.....	83
7.	Revision History	84

1. CloudStream Roles

CloudStream contains three levels of user provisioning. Each user is assigned one of the following roles and has varying access to the **CloudStream System Control**:

Role	Description	Created By
System	The System user is only used during the initial installation and provisioning of CloudStream. Only certified CloudStream installers may access this account. Once initial installation and configuration is complete, access to this account is revoked. The system account is capable of creating new CloudStream nodes and removing CloudStream nodes.	• CloudStream Installer
Admin	Admin users can perform all levels of CloudStream operations within the CloudStream System Control including: • Create/Remove other users/admins • Change the role of other users/admins • Update the password of other users/admins • Deploy Cloudstream to a node (Updating the node version) • Pause/Resume operations on a Node • Create/Delete/Edit CloudStream Sources • Create/Delete/Edit CloudStream Destination • Create/Delete/Edit CloudStream Services • View Logs	• CloudStream Installer • Admin
User	Users can perform all basic CloudStream operations and have limited user administration capabilities within the CloudStream System Control including: • Update password • Pause/Resume operations on a Node • Create/Delete/Edit CloudStream Sources • Create/Delete/Edit CloudStream Destination • Create/Delete/Edit CloudStream Services • View Logs	• Admin

After installation, your certified installer will provide you with an initial Admin account. Depending on your installation type, you will either receive a username/password (Edge Only: typically "admin"/"admin"), or you will receive an email invite to join CloudStream Cloud.

For additional details on user creation and administration, see **2.5 User Management**.

2. CloudStream System Control

All user operations of CloudStream take place within the **CloudStream System Control** web application. Depending on your installation, the web application will be hosted in the cloud or locally at the edge.

2.1. Installation

Note: Installation of CloudStream must be performed by a certified installer and the type of installation is determined during licensing. Future versions of CloudStream will allow for self-performed installs.

A CloudStream installation consists of at least 1 CloudStream Controller and 1 CloudStream Node. CloudStream is a fully distributed system, comprised of a 1 controller: n node model. A Controller can control any number of Nodes. Controllers and Nodes can be installed in any combination of Edge and Cloud. Your installer will work with you to determine the optimal installation for your use case.

The following environments are supported for installation:

Cloud: Casne Hosted Cloud

Edge Containerized (Docker): Linux, Windows 10, Windows 11, Windows Server 2022

Edge Non-Containerized: Windows 10, Windows 11, Windows Server 2016, Windows Server 2019, Windows Server 2022.

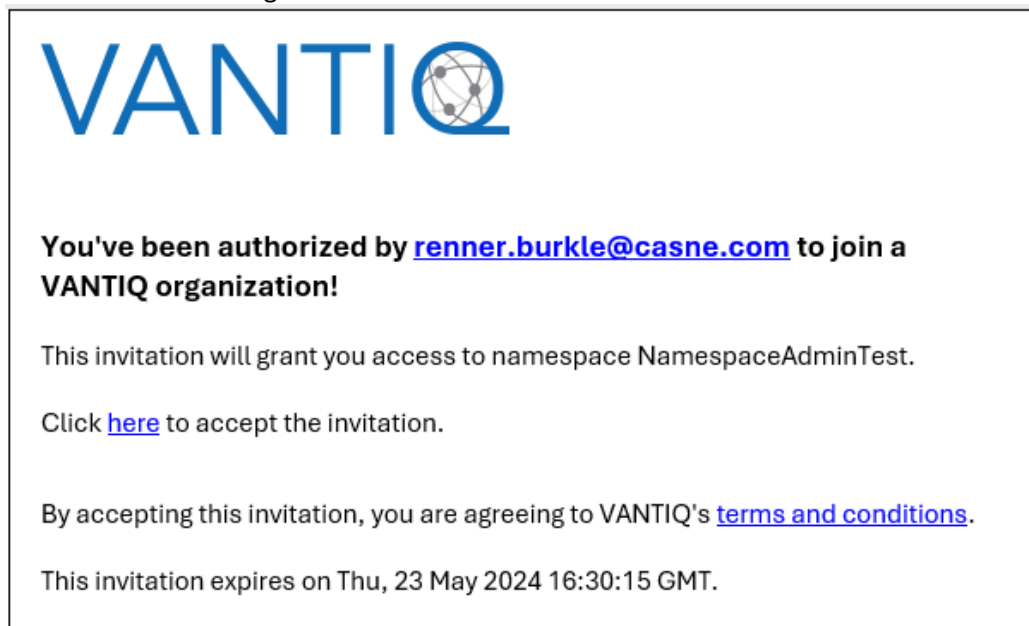
For Edge systems the minimum recommended resources are 8GB of RAM and 4CPU cores.
Depending on data transfer rates, additional resources may be required.

2.2. System Control Access

Access to the CloudStream System Control web application is dependent on the type of installation. CloudStream System Control is bundled with every CloudStream Controller instance.

Cloud Hosted Controllers

You will receive an email invite from operations@vantiq.com containing a link to your CloudStream cloud organization.



Edge Hosted Controllers

Your CloudStream installer will provide you with a URL (typically localhost:8080 or [hostname]:8080) after installation, along with an initial admin account (typically "admin"/"admin"). Navigate to the URL to access CloudStream System Control.

2.3. Node Management

The “Nodes” page is the initial page all users are directed to upon logging in to CloudStream system control. Management of CloudStream nodes (including creating/deleting nodes, deploying/updating CloudStream versions, and updating the data services running on a node) is performed on this page. All other pages are locked until a node has been selected to manage.

The screenshot displays the CASNE CloudStream System Control interface. On the left is a dark sidebar with navigation links: Nodes (selected), Config, Logs, System, Users, and Logout. The main content area has a header with the CASNE logo, 'CloudStream System Control', and 'Version 2.0'. Below the header are three buttons: 'Create Node', 'Deploy CloudStream', and 'Refresh'. A table lists the nodes, with one entry visible: 'RJB_2_0_Node' with version 2.0, created on 2024-08-22 16:22:21, and a latency of 189 ms. The table has columns for Name, Node Version, Creation Time, Deployed?, Latency (ms), Config, and Delete. The 'Deployed?' column shows 'SUCCESS'. The 'Config' column has a 'Manage' link, and the 'Delete' column has a 'Delete' link. At the bottom right of the table are 'Back', '1', and 'Next' buttons.

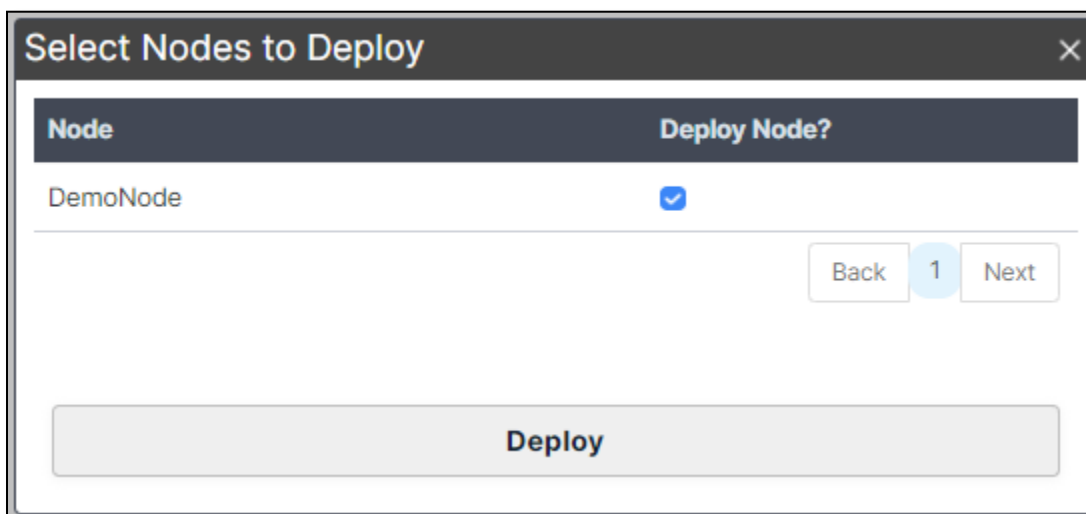
Name	Node Version	Creation Time	Deployed?	Latency (ms)	Config	Delete
RJB_2_0_Node	2.0	2024-08-22 16:22:21	SUCCESS	189	Manage	Delete

Node Creation

Node creation activities are restricted to the **System** (installation) user, and all nodes are created during the installation process. New nodes are created by clicking the “Create Node” button. This button is disabled for admin and user leveled users.

Deploy CloudStream

The “Deploy CloudStream” button is used to initially deploy CloudStream to a node, or to upgrade a node with the latest version of the CloudStream software. The CloudStream software is contained with the CloudStream Controller instance and can be remotely deployed to any connected nodes. Clicking this button opens the node deployment popup and users may select which nodes to deploy to:



The dialog box titled "Select Nodes to Deploy" contains a table with two columns: "Node" and "Deploy Node?". The table has one row with "DemoNode" in the first column and a checked checkbox in the second column. Below the table, there are "Back", "1", and "Next" buttons. At the bottom of the dialog is a large "Deploy" button.

Node	Deploy Node?
DemoNode	☒

Back 1 Next

Deploy

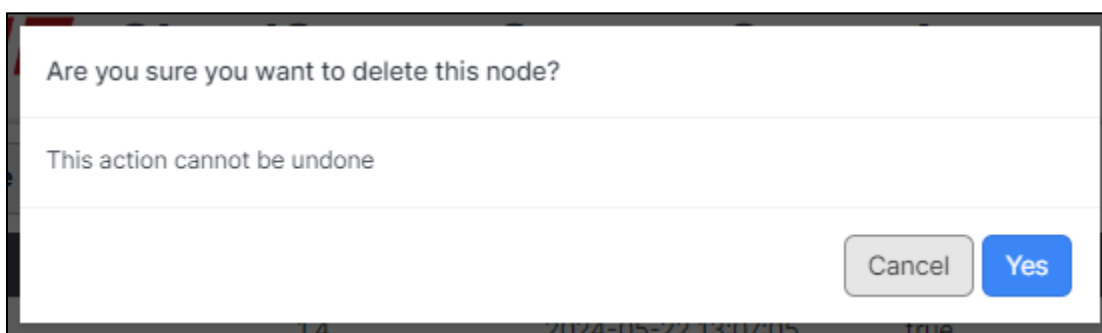
Depending upon connection speeds, deployment of the CloudStream software may take up to 10 minutes. Deployment status can be checked at any time by pressing the "Refresh" button. Once the deployment status of a node is "true", a node may be managed. The "Deploy CloudStream" button is disabled for user leveled users.

Node Config

Clicking the "Manage" button for a node will navigate users to the Config page ([see section 2.4](#)). Once a node has been selected, Config, Logs, and System pages are unlocked.

Node Deletion

Clicking the "Delete" button will permanently delete a node. This button is disabled for admin and user leveled users. Users are asked to confirm their selection prior to deleting a node:



The dialog box asks "Are you sure you want to delete this node?" and states "This action cannot be undone". It has "Cancel" and "Yes" buttons at the bottom right. At the bottom of the dialog, there is a status bar showing "14", "2024-05-22 13:07:05", and "true".

Are you sure you want to delete this node?

This action cannot be undone

Cancel Yes

14 2024-05-22 13:07:05 true

2.4. Node Configuration (“Config”)

All node configuration activities take place within the Config page. From this page, users can create [Data Sources](#), [Data Destinations](#), [Data Injectors](#), and [CloudStream Services](#).

The modular building blocks of data operations in CloudStream are Sources, Destinations, Injectors and CloudStream Services. Once sources and destinations have been defined, data is routed between them via CloudStream services. Data Injectors modify data in real-time, before the data is sent to the destination.

The screenshot displays the CASNE CloudStream System Control interface. The left sidebar contains navigation links: Nodes, Selected Node 2_0Node, Node State Running, Config (highlighted), Logs, System, Users, and Logout. The main content area is titled "CASNE CloudStream System Control" and "2_0Node". It features three tabs: Sources, Injectors, and Destinations. The Sources tab is active, showing a "New Source" button and a table with columns: Name, Type, Rate (sec), Data Format(s), and Status. The table contains one entry: "PI-EventFrames" (PI Web API Event Frames) with a rate of 10 and data format of JSON Object, status "Active". The right panel shows "CloudStream Services" with a "New Service" button and a table with columns: Name, Source, Destination, Injectors, Status, and Trend. The table lists two services: "EventFrame→MQTT" (PI-EventFrames to Hive-out) and "EventFrame→CDS" (PI-EventFrames to CDS-Event-Out), both with 0 injectors and "Active" status. Below the services table is an "Event Counts" section with a "Live Data" dropdown and a message: "No Data Displayed. Check 'Trend' for a Service to display on the graph."

2.5. User Management (“Users”)

Edge Deployments

Each Edge deployment of CloudStream is initialized with a single ‘Admin’ typed user.

Using the **Users** page, ‘Admin’ typed users can:

- Add new users (‘Admin’ or ‘User’ typed) at any time by clicking the **Add User** button.
- Update the user type or password of any user by clicking the **Update** button.
- Remove other users by pressing the **Remove** button.

Note: If there is only 1 user remaining in the user list, it cannot be deleted. If the only remaining ‘Admin’ typed user is somehow removed, please contact Casne Engineering.

‘Admin’ and ‘User’ typed users may change their password at any time by clicking on the Users page and submitting a new password.

The screenshot shows the CASNE CloudStream User Control interface. On the left is a sidebar with navigation links: Nodes, Config, Logs, System, Users, and Logout. The main content area has a header with the CASNE logo and 'CloudStream User Control'. Below the header, there are two main sections. The 'Current User' section on the left shows a text field with 'system' and an 'Update Password' section with 'New Password' and 'Confirm New Password' fields, each with a red eye icon, and a 'Submit' button. The 'All Users' section on the right has an 'Add User' button and a table with columns 'Username', 'User Type', and 'Created By'. The table contains two rows: 'admin' (Admin, system) and 'cloudstreamUser' (User, system). Each row has 'Update' and 'Remove' buttons. At the bottom of the table are 'Back', '1', and 'Next' buttons.

This screenshot shows the same CASNE CloudStream User Control interface as the previous one, but with a 'Create new user' modal window open in the center. The modal has a title bar with a close button. It contains fields for 'Username', 'User Type' (with radio buttons for 'User' and 'Admin'), 'Password', and 'Password (confirm)', each with a red eye icon. There is a 'Submit' button at the bottom of the modal. The background interface is dimmed.

Cloud Deployments

'Admin' typed users cannot add new users to managed Cloud Deployments. New users are added by Casne Engineering. Please send new user requests to CloudStream@casne.com.

'Admin' and 'User' typed users may change their password at any time by clicking on the Users page and submitting a new password.

CASNE CloudStream User Control

Current User:

renner.burkle@casne.com

Update Password

New Password

Confirm New Password

Submit

All Users:

Add User

Username	User Type	Created By	
renner.burkle@casne.com	Admin	CloudStream Admin	Update Remove

Back 1 Next

2.6. CloudStream System Control (“System”)

The System page contains CloudStream controls for the selected node. This includes starting/stopping the CloudStream clock or resetting CloudStream’s internal memory. This page also includes controls for the Data Injectors Subservice.

2.6.1. CloudStream Control

CloudStream Control

Start Clock

Resume Services

Stop Clock

Pause All Services

Reset Memory

Clear Internal Memory

Clock Active	Last Modified
true	2024-08-27 11:03:54

Start Clock

Many operations within CloudStream are clock driven, such as polling from Data Sources. These operations will be paused when the Clock is stopped. However, some data sources (for example: MQTT), push data directly to CloudStream and are unaffected by the system clock status. The clock starts in a stopped state for new CloudStream nodes and must be started here prior to node operation. The clock status is also displayed as the “Node State” in the navigation bar once a node has been selected.

Stop Clock

Clicking this button will stop the CloudStream clock, which will pause any CloudStream services that utilize the system clock.

Reset Memory

Clicking this button will reset the internal memory utilized by CloudStream. CloudStream caches certain data (such as PI Web API WebIds and schemas for AF Elements) in order to optimize latency and remote data calls. Resetting the internal memory will force CloudStream to re-cache this data and may be used to immediately pick up changes made against external data systems.

2.6.2. Data Injector Subservice Management (Edge Deployments)

Data Injectors Subservice Control

Execution Source

Edge

Subservice Status

Running

Refresh

Advanced:

Show Token

Import Modules

Execution Source

Indicator for the Subservice deployment location – “Edge” or “Cloud”.

Subservice Status

The current status of the Injector Subservice. The status will either be: (1) **“Running”** – the subservice is in a healthy state and ready to perform data injections, (2) **“Uninitialized”** – the service has not been created, or (3) **“Inactive”** – the service has been created but is in an unhealthy or stopped state and unable to perform data injections.

Note: The status is rechecked every time this page is loaded and may be manually updated by pressing the “Refresh” button. If the Cloud injector subservice is in an unhealthy state, it can be rebooted by restarting the Data Injectors Docker Container or Windows Task (depending on deployment). Please wait for up to 10 seconds for the service to completely reboot.

Show Token

The “Show Token” button generates an authentication token for the subservice to securely connect to the CloudStream Node. This token is only used by CloudStream installers during subservice configuration.

Import Modules

Clicking this button will launch the module import interface. Use this interface to install python packages or modules for use in data injector scripts. This interface will also show all currently installed modules and packages

Import Python Modules or Packages

Enter modules and versions to install, or import a requirements.txt file.
Modules should be in the form 'name[==version]', for example: 'numpy' or 'numpy==1.25.1'

Import Requirements.txt

Modules to be imported

Currently installed modules

- aiohttp=3.9.5
- aiosignal=1.3.1
- attrs=23.2.0
- codetiming=1.4.0
- frozenset=1.4.1
- idna=3.7
- jprops=2.0.2
- multidict=6.0.5
- pip=24.0
- psycogp2=2.9.9
- setuptools=69.0.3
- vantiqconnectorsdk=1.2.3
- vantiqPythonExecConnector=1.2.3
- vantiqsdk=1.2.2

Submit

You may import a requirements.txt file, or manually enter a list of modules/packages to install. Submitted the list will send the installation requests to the subservice and a log message will be generated for each installation request.

Note: The injector subservice will be unavailable for data injections while modules/packages are installing. Larger packages may take up to 10 minutes.

You may confirm which packages have been installed by returning to the module import interface and checking the currently installed modules list.

2.6.3. Data Injector Subservice Management (Cloud Deployments)

Data Injectors Subservice Control

Execution Source

Cloud

Subservice Status

Running

Refresh

Reboot

Advanced:

Register Node

Connect Node

Disconnect Node

Import Modules

Customer	Node Connected	License Status	License End Date	Injectors Deployed	Total Purchased
CasneEngineering	true	Active	2124-08-01 17:24:14	1	99

Execution Source

Indicator for the Subservice deployment location – “Edge” or “Cloud”.

Subservice Status

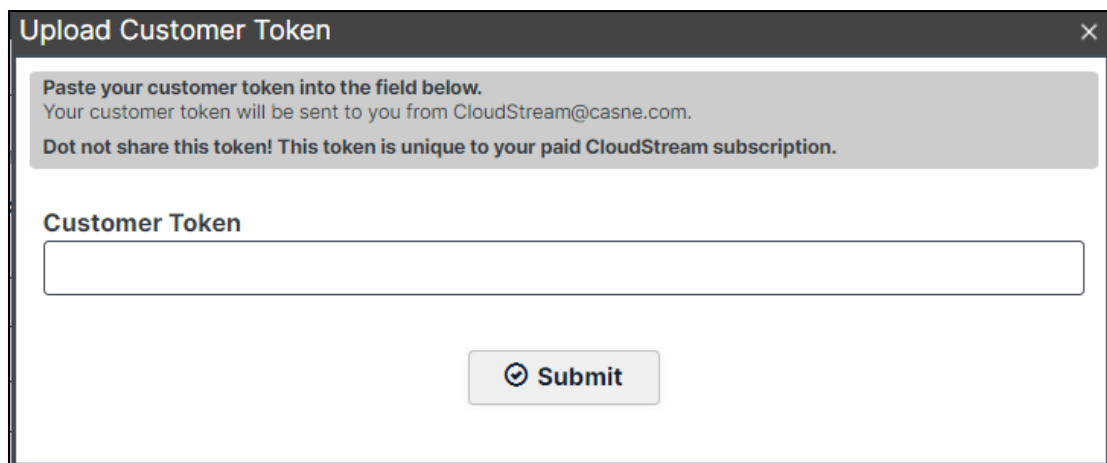
The current status of the Injector Subservice. The status will either be: (1) **“Running”** – the subservice is in a healthy state and ready to perform data injections, (2) **“Uninitialized”** – the service has not been created, or (3) **“Inactive”** – the service has been created but is in an unhealthy or stopped state and unable to perform data injections.

Note: The status is rechecked every time this page is loaded and may be manually updated by pressing the “Refresh” button. If the managed Cloud injector subservice is in an unhealthy state, it can be rebooted by pressing the “Reboot” button. Please wait for up to 5 minutes for the service to completely reboot. If your injector subservice remains in an unhealthy state, please contact CloudStream@casne.com.

Advanced Controls:

Register Node

Register your node using your customer token. All managed cloud data injector customers are emailed a customer token upon purchasing Casne CloudStream. Entering your token into the pop-up will register your node to your data injectors subscription. Once registered, you may connect a data injector subservice to your node.



The screenshot shows a dialog box titled "Upload Customer Token" with a close button (X) in the top right corner. Inside the dialog, there is a greyed-out instruction box that reads: "Paste your customer token into the field below. Your customer token will be sent to you from CloudStream@casne.com. Dot not share this token! This token is unique to your paid CloudStream subscription." Below this, the label "Customer Token" is positioned above a text input field. At the bottom center of the dialog is a button with a checkmark icon and the text "Submit".

Connect Node

Clicking this button will create a new managed cloud instance of the Data Injectors subservice. The subservice will automatically connect to your node. This will fail if your license is inactive, your license has expired, or if you have already provisioned the maximum amount of injector subservices purchased.

Disconnect Node

Clicking this button will disconnect and delete the managed cloud instance of the Data Injectors subservice. The number of injectors deployed will decrease by 1, and you may connect this service to a different node.

Import Modules

Clicking this button will launch the module import interface. Use this interface to install python packages or modules for use in data injector scripts. This interface will also show all currently installed modules and packages

Import Python Modules or Packages

Enter modules and versions to install, or import a requirements.txt file.
Modules should be in the form 'name[==version]', for example: 'numpy' or 'numpy==1.25.1'

Import Requirements.txt

Modules to be imported

Currently installed modules

- aiohttp=3.9.5
- aiosignal=1.3.1
- attrs=23.2.0
- codetiming=1.4.0
- frozendict=1.4.1
- idna=3.7
- jprops=2.0.2
- multidict=6.0.5
- pip=24.0
- psycpg2=2.9.9
- setuptools=69.0.3
- vantiqconnectorsdk=1.2.3
- vantiqPythonExecConnector=1.2.3
- vantiqsdk=1.2.2

Submit

You may import a requirements.txt file, or manually enter a list of modules/packages to install. Submitted the list will send the installation requests to the subservice and a log message will be generated for each installation request.

Note: The injector subservice will be unavailable for data injections while modules/packages are installing. Larger packages may take up to 10 minutes.

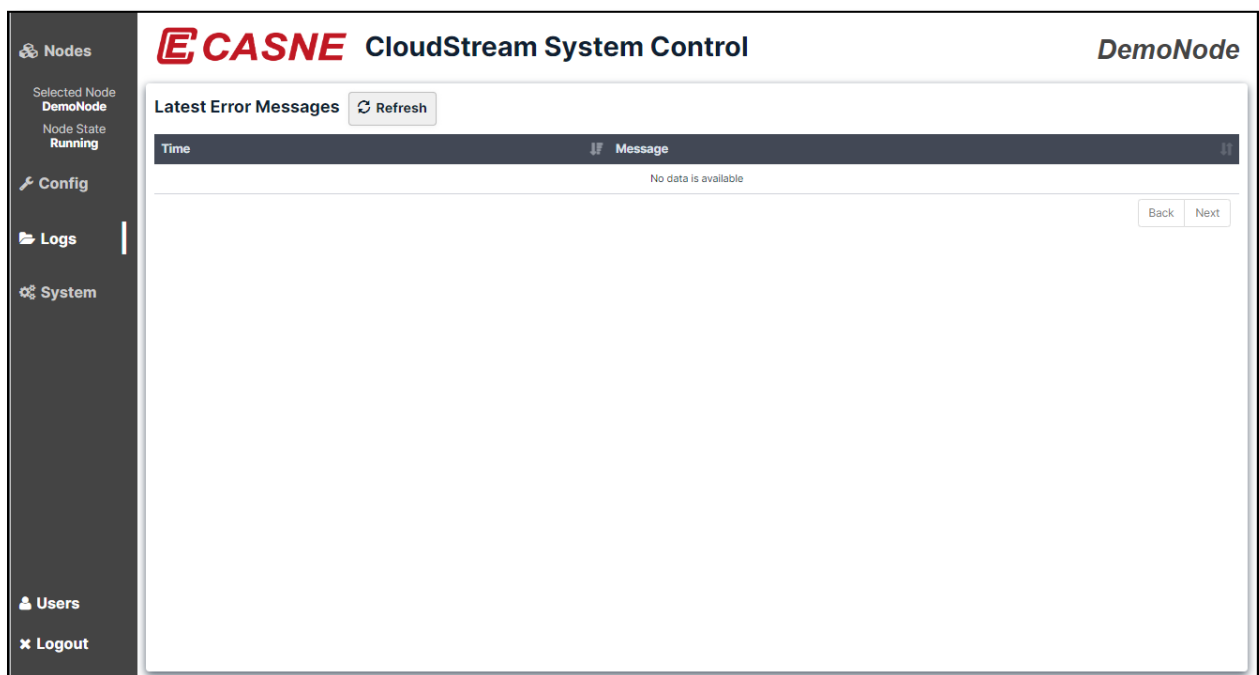
You may confirm which packages have been installed by returning to the module import interface and checking the currently installed modules list.

2.7. Logging (“Logs”)

The Logs page contains all error messages generated during CloudStream node operation. These may include errors from accessing source systems, destination systems, or internal CloudStream errors.

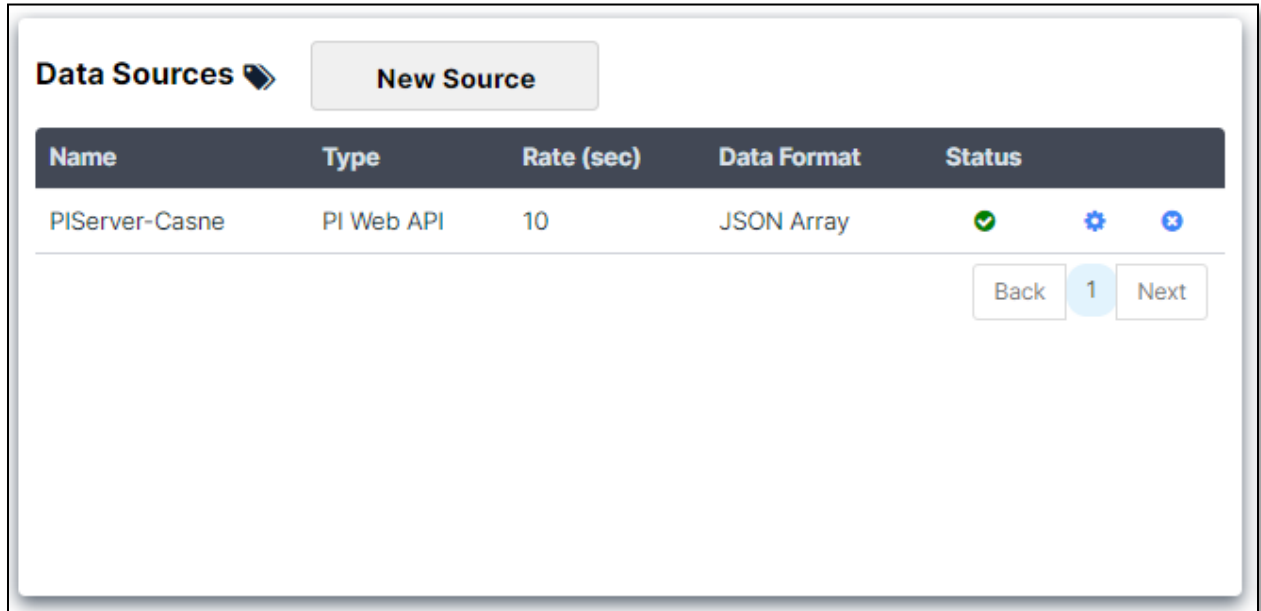
The logs refresh automatically upon loading the page and may be refreshed at any time by pressing the “Refresh” button.

Note: A maximum of 1,000 log messages are kept at any given time. Logs do not have an expiration date. Once 1,000 log messages have been received the oldest logs are automatically discarded once new logs arrive.

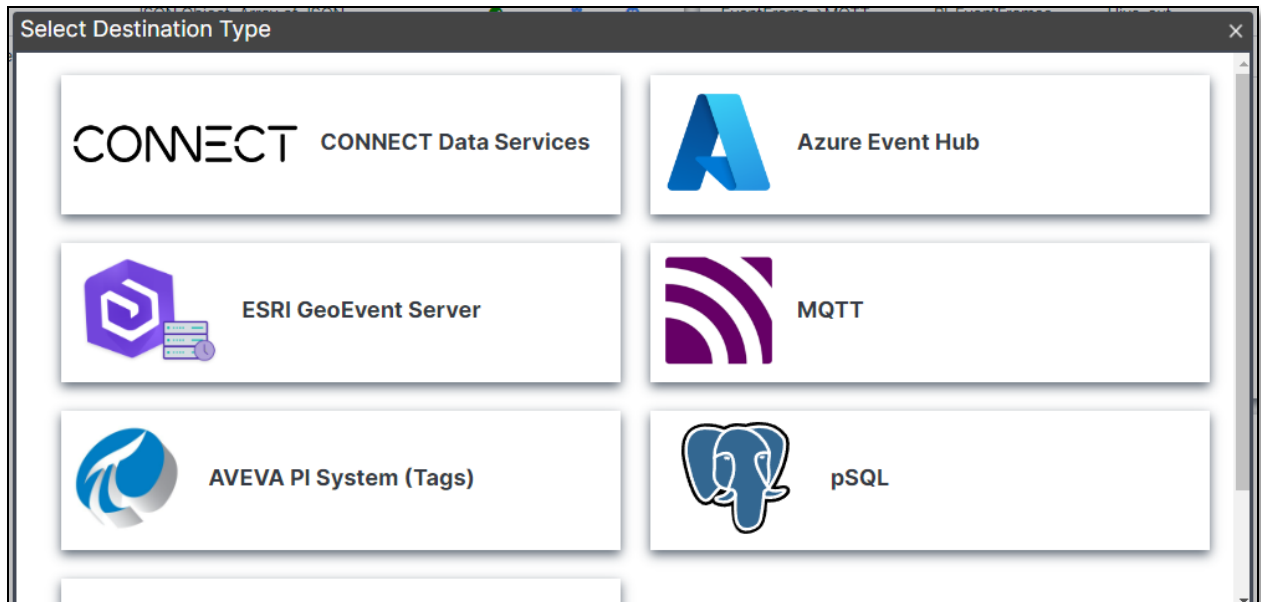


3. CloudStream Source Creation

To create a new CloudStream source, click on the “**New Source**” button from the Config page. Existing sources may be edited by pressing the gear button. Sources may be deleted by pressing the delete button (Note: a Source cannot be deleted if it is part of a Service).



This will open the Source Selection pop-up:



Click on any source to open up the corresponding Source Configuration window.

3.1.AVEVA CONNECT Data Services

The AVEVA CONNECT Data Services Source polls data from AVEVA CONNECT Data Services SDS Streams.

Tab 1: Connection Info

Configure New Data Source

Connection Info

Enter Name, Connection Info & Validate to Continue.

CloudStream Data Source Name

Ex: CONNECT Data Service 1

Client ID

Please enter your CONNECT Data Service Client ID

Client Secret

Please enter your CONNECT Data Service Client Secret

Select Region:

Select Region

Validate Connection

Data Preview

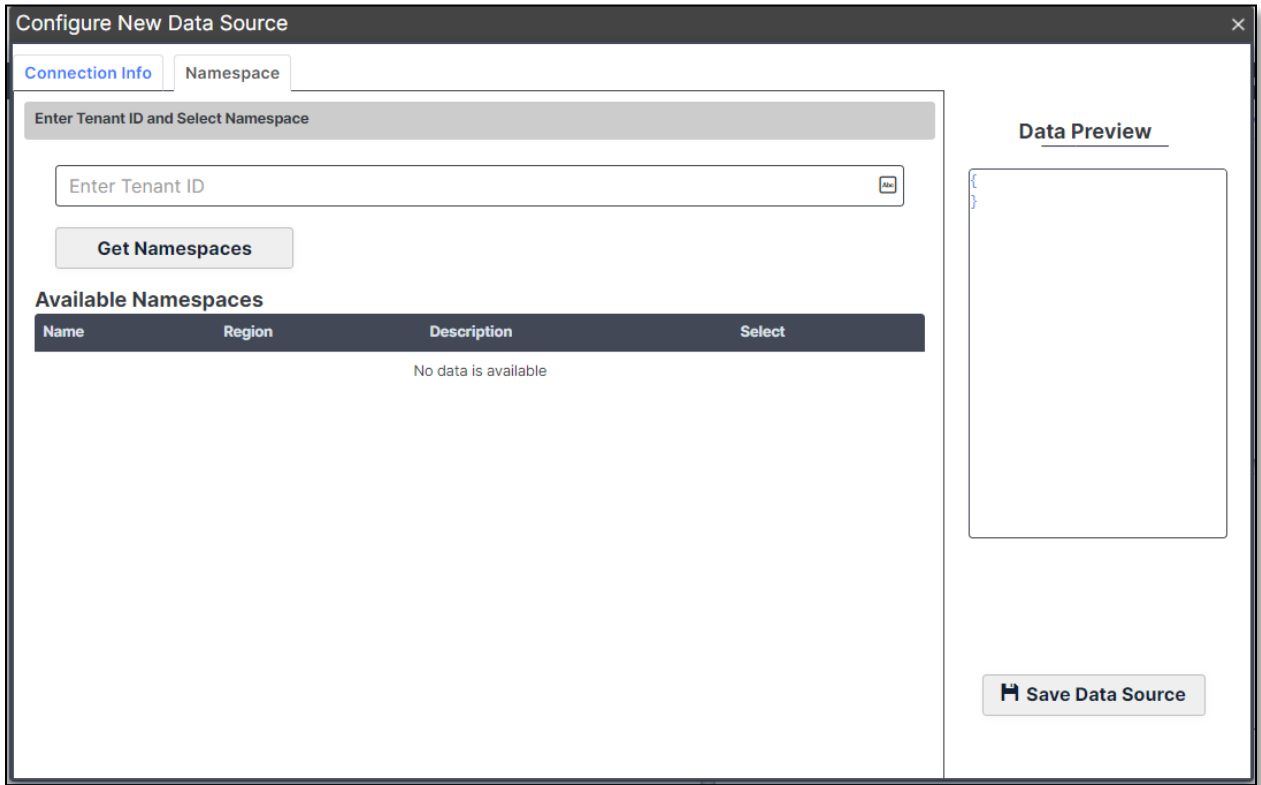
{ }

Save Data Source

Instructions for this tab:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter your AVEVA CONNECT Data Services **Client ID** and **Client Secret**.
3. Select your Data Hub Region (Currently supported: **US-West**)
4. Click "Validate Connection".
 - a. If successful, you will automatically move to the next configuration step ("**Namespace**" Tab).
 - b. In unsuccessful, you will be given an error message. Please correct the **Client ID**, **Client Secret**, or **Region**.
 - c. A Data Source may be saved at any time once Validation is complete.

Tab 2: Namespace



Configure New Data Source

Connection Info | **Namespace**

Enter Tenant ID and Select Namespace

Enter Tenant ID

Get Namespaces

Available Namespaces

Name	Region	Description	Select
No data is available			

Data Preview

```
{  
}
```

Save Data Source

Instructions for this tab:

1. Enter your Tenant ID and click **"Get Namespaces"**.
2. Select the Namespace containing the Streams you want to poll data from.
 - a. Only 1 Namespace can be selected per CloudStream AVEVA CONNECT Data Services Source
 - b. Once you select a **Namespace** you will automatically navigate to the **"Streams"** tab.

Tab 3: Streams

Configure New Data Source

Connection Info

Namespace

Streams

Select Streams to Query

CloudStream will poll from all selected streams based on the chosen Rate:

Data Re-Check Rate (seconds):
10

Available Streams

Name	Typeld	Select
Test1	TEST	▶

Selected Streams

Name	Typeld	Remove
No data is available		

Data Preview

{
}

Save Data Source

Instructions for this tab:

1. Enter your desired **Data Re-Check Rate**.
 - a. The minimum re-check rate is 1 second.
 - b. This is the rate at which CloudStream checks for new data.
 - c. For data that updates faster than your re-check rate, CloudStream will receive all updated values since the last check. No data is lost for faster updating data streams.
2. Select Streams you want to poll data from by pressing the button under the “**Select**” tab.
 - a. You may select multiple streams, but they must have the same **Typeld**.
 - b. An error will be generated if Streams with different **Typelds** are selected.
 - c. Upon selecting a Stream, the Data Preview will automatically update with the selected **Typeld**.
3. You may de-select a stream by pressing the button under the “**Remove**” column.
4. Once at least one Stream is selected, you may move to the “**Transforms**” tab.

Tab 4: Transforms

Configure New Data Source

Connection Info | **Namespace** | Streams | Transforms

(Optional) Alter the outgoing Data Object from AVEVA Data Hub
Preview and confirm the outgoing Data Object before saving.

Add or Remove Properties from the Data Object

You may add additional properties with static values, or remove properties.

{"timestamp":16," } Click to expand

Override Property Names

Enter overrides in the form of:
`{"[Original Property]" : "[New Name]"}`
For example: {"Value" : "Pressure01"}

You can dynamically assign the name of the Stream to a property using @StreamName. For example: {"Value" : "@StreamName"}

{"Value":"Value"} Click to expand

Data Preview

```
{
  timestamp: 16,
  value: 14
}
```

Instructions for this tab:

1. (Optional) Add or Remove Properties from the Data Object:
 - a. Click on the object symbol { } to open the object editor.
 - b. Unwanted properties can be deleted. CloudStream will not send deleted properties.
 - c. Additional, static properties can be added. Enter a new property name and the static value for that property.
2. (Optional) Override Property Names
 - a. Click on the object symbol { } to open the object editor.
 - b. Override property names by entering the old property name on the left, and the new property name on the right.
 - i. For example, "timestamp":"Time" will rename the "timestamp" property to "Time" in the outgoing data message.
 - ii. Entering the value of "@StreamName" on the right will dynamically rename the property to the name of the SDS stream from AVEVA CONNECT Data Services.
 1. This is useful when polling data from multiple streams from the same CloudStream Source.
 2. Note: If the SDS Stream Name is not a valid JSON property name, this transform will fail.
3. When complete, **Save** the Data Source and exit the configuration pop-up.

3.2. CSV

The CSV Source reads and sends data from an uploaded CSV file.

Note: CSV Source sends data all at once.

Tab 1: CSV Upload

Configure New Data Source

CSV Upload

Upload CSV & Validate Data Preview.
For best results, do not exceed 10,000 records per CSV.

CloudStream Data Source Name

Ex: CSV Data

Upload CSV

Uploaded Document Name

Data Preview

Save Data Source

Instructions for this tab:

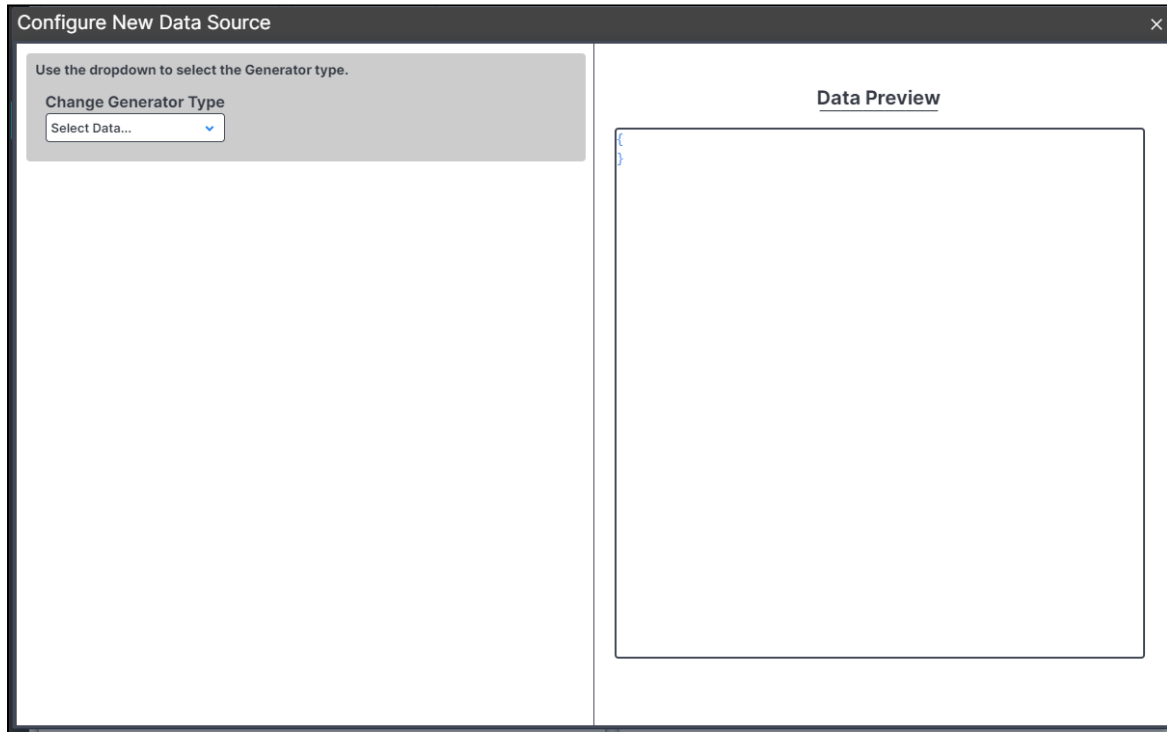
1. Enter a name for the CloudStream Data Source into the top input box.
2. Click "Upload CSV"
 - a. Select a CSV file that you would like to make as a data source
 - b. If successful, you will be given a message "File has been uploaded to document 'XXXX'"
 - c. If unsuccessful, you will be given an error message.

Note: Users will see a preview of the Data in the Data Preview section

3. When complete, **Save** the Data Source and exit the configuration pop-up.

3.3. Data Generator

The Data Generator Source simulates data using a variety of methods, including CSV playback, Sinusoid, Random and Totalizer algorithms.



Instructions for this Data Source:

1. Select a **Generator Type** from the dropdown, then enter configuration items for the selected algorithm.

Generator Type: CSV File

The CSV File generator continuously plays back data from an uploaded CSV file. When the final record from the CSV is sent, playback restarts from the first record.

The screenshot shows a 'Configure New Data Source' window with a dark title bar and a close button. The window is divided into two main sections. The left section contains configuration options: a grey box at the top with instructions to 'Upload CSV & Validate Data Preview' (not exceeding 10,000 records), a 'Change Generator Type' dropdown menu set to 'CSV File', a 'Data Source Name' input field with the example 'Ex: CSV Data', a 'Data Playback Rate (Sec)' input field with the example 'Ex: 30', an 'Upload CSV' button, and an 'Uploaded Document Name' input field. At the bottom left is a 'Save Data Source' button. The right section is titled 'Data Preview' and contains a large, empty rectangular box for displaying data.

Instructions for this Data Source:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter the speed at which data is replayed "Data Playback Rate". A single record will be sent from the CSV every time the interval has been met.
3. Click "Upload CSV"
 - a. Select a CSV file using the file browser popup.
 - b. If successful, users will receive the message "File has been uploaded to document 'XXXX'"
 - c. If unsuccessful, users will be given an error message.

Note: After uploading the file users will see a preview of the Data in the Data Preview section

When complete, **Save** the Data Source and exit the configuration pop-up.

Generator Type: **Sinusoid**

The sinusoid generator generates data using the following sinusoid algorithm:

```
value = Amplitude*(Math.sin(((Rate*i)*2*Math.PI()/Period)))
```

Where:

Amplitude: The maximum value for the sine function

Rate: The rate in which data is generated

Period: The duration of time for 1 complete sine wave.

i: A counter for the instance in which rate has been met. Starts at 0 and increments by 1 each time data is generated and resets when (rate * i) exceeds the period.

Every message is sent in the format of:

```
{
  "value": [generated value],
  "timestamp": [timestamp with time zone (UTC) when CloudStream generated the event]
}
```

Configure New Data Source

Choose Amplitude, Period, & Rate
Use the dropdown to change Generator type.

Change Generator Type
Sinusoid

Data Source Name
mySin

Amplitude
100

Period (seconds)
120

Rate (seconds)
10

Save Data Source

Data Preview

```
{
  0: {
    Timestamp: 1,
    Value: 0
  },
  1: {
    Timestamp: 2,
    Value: 50
  },
  2: {
    Timestamp: 3,
    Value: 86.6025
  },
  3: {
    Timestamp: 4,
    Value: 100
  },
  4: {
    Timestamp: 5,
    Value: 86.6025
  },
  5: {
    Timestamp: 6,
    Value: 50
  },
  6: {
    Timestamp: 7,
    Value: 0
  }
}
```

Instructions for this Data Source:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter a number for the **Amplitude** into the input box.
3. Enter a number for the **Period** into the input box.
4. Enter a number for the **Rate** into the input box.
5. When complete, Save the Data Source and exit the configuration pop-up.

Generator Type: **Random**

The Random generator generates data using the following algorithm:

```
value = Math.random(LowerBound, UpperBound)
```

Where:

UpperBound: The maximum value for the random function

LowerBound: The minimum value for the random function

Every message is sent in the format of:

```
{
  "value": [generated value],
  "timestamp": [timestamp with time zone (UTC) when CloudStream generated the event]
}
```

The screenshot shows a 'Configure New Data Source' window with two main sections: configuration on the left and a data preview on the right.

Configuration Section:

- Choose Upper and Lower Bounds & Rate:** A sub-header with the instruction 'Use the dropdown to change Generator type.'
- Change Generator Type:** A dropdown menu currently set to 'Random'.
- Data Source Name:** A text input field containing 'myRandom'.
- Upper Bound:** A text input field containing '100'.
- Lower Bound:** A text input field containing '100'.
- Rate (seconds):** A text input field containing '10'.
- Save Data Source:** A button at the bottom left.

Data Preview Section:

- Data Preview:** A sub-header above a code editor.
- Code Editor:** Displays a JSON array of 7 objects. Each object has a 'Timestamp' and a 'Value' of 100. The timestamps are 1 through 7.

Instructions for this Data Source:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter a number for the **Upper Bound** into the input box.
3. Enter a number for the **Lower Bound** into the input box.
4. Enter a number for the **Rate** into the input box. (A message will be sent every X seconds, where X is the Rate).
5. When complete, Save the Data Source and exit the configuration pop-up.

Generator Type: **Totalizer**

The Totalizer generator generates data using the following sinusoid algorithm:

```
value = Peak*[i*(rate/frequency) - math.floor(i*(rate/period))]
```

Where:

Peak: The maximum value for the totalizer

Rate: The rate (seconds) in which data is generated

Frequency: The frequency (seconds) in which the totalizer resets

i: A counter for the instance in which rate has been met. Starts at 0 and increments by 1 each time data is generated and resets when (rate * i) exceeds the frequency.

Every message is sent in the format of:

```
{
  "value": [generated value],
  "timestamp": [timestamp with time zone (UTC) when CloudStream generated the event]
}
```

Note that whenever the totalizer resets, 2 data points are set with an offset of 1ms: the final point (value = peak), and the reset point (value = 0).

Configure New Data Source

Choose Peak, Frequency & Rate
Use the dropdown to change Generator type.

Change Generator Type
Totalizer

Data Source Name
myTotalizer

Peak
100

Frequency (seconds)
120

Rate (seconds)
10

Save Data Source

Data Preview

```
{
  0: {
    Timestamp: 1,
    Value: 0
  },
  1: {
    Timestamp: 2,
    Value: 8.3333
  },
  2: {
    Timestamp: 3,
    Value: 16.6667
  },
  3: {
    Timestamp: 4,
    Value: 25
  },
  4: {
    Timestamp: 5,
    Value: 33.3333
  },
  5: {
    Timestamp: 6,
    Value: 41.6667
  },
  6: {
    Timestamp: 7,
    Value: 7
  }
}
```

Instructions for this Data Source:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter a number for the **Peak** into the input box.
3. Enter a number for the **Frequency** into the input box.

4. Enter a number for the **Rate** into the input box.
5. When complete, Save the Data Source and exit the configuration pop-up.

3.4. MQTT

The MQTT source subscribes to MQTT topics and receives incoming data into CloudStream. This source does not utilize the CloudStream clock, and messages are received in real time as they are sent to the broker.

Tab 1: Connection Info

Configure New Data Source

Connection Info

Enter Name, Connection Info & Validate to Continue.

CloudStream Data Source Name

Ex: MQTT-in

Connection

Broker URL

Ex: localhost, mqtt.dev

Broker Port

Ex: 1883

Broker Encryption

☒ Unencrypted (tcp)

☐ Encrypted (tcps)

QoS

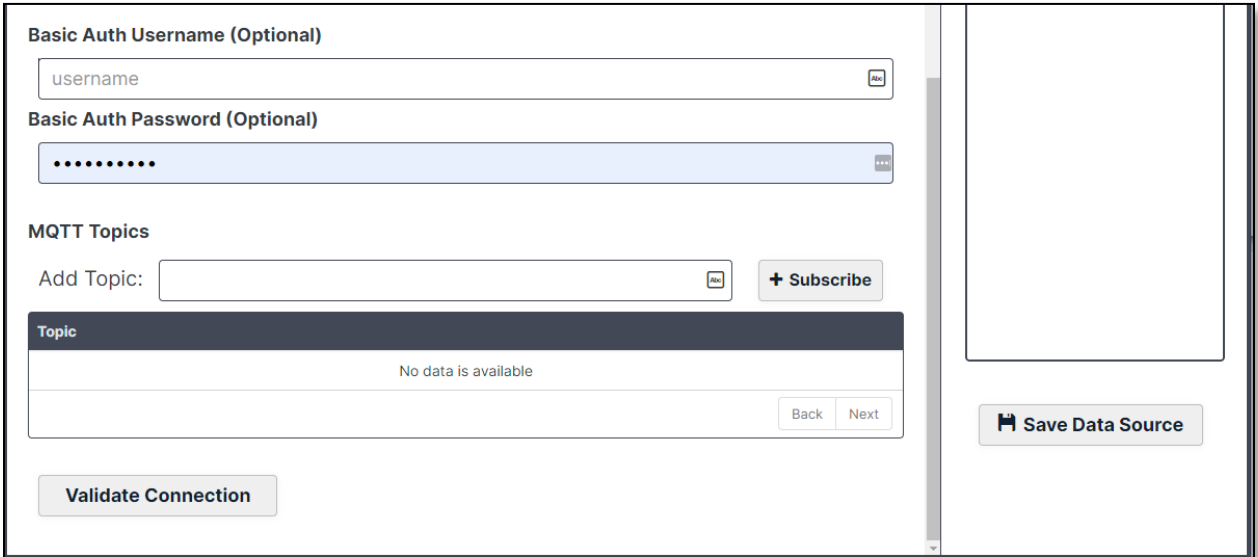
QoS 0

Data Preview

Not Available

Instructions for this tab (Part 1):

1. Enter the name for the CloudStream Data Source into the top input box.
2. Enter the server name and port for the MQTT broker.
3. Choose encryption mode for the broker (TCP or TCPS)
4. Select a QoS (default QoS is 0)
 - QoS 0 (At most once) provides a best-effort delivery system where the sender does not wait for an acknowledgment or guarantee delivery. The recipient does not confirm receipt, and the sender neither stores nor retries sending the message.
 - QoS 1 (At least once) ensures that the message is delivered to the receiver at least once. When a message is published with QoS 1, the sender retains a copy of the message until it receives a PUBACK packet from the receiver, confirming successful delivery.
 - QoS 2 (Exactly once) offers the highest level of service in MQTT, ensuring that each message is delivered exactly once. This is achieved through a four-step handshake process between the sender and the receiver.



The screenshot shows a web interface for configuring a data source. It includes fields for 'Basic Auth Username (Optional)' and 'Basic Auth Password (Optional)'. Below these is a section for 'MQTT Topics' with an 'Add Topic' input field and a '+ Subscribe' button. A table with the header 'Topic' is shown, containing the text 'No data is available'. At the bottom of the main form area is a 'Validate Connection' button. On the right side, there is a large empty box and a 'Save Data Source' button at the bottom.

Basic Auth Username (Optional)

username

Basic Auth Password (Optional)

.....

MQTT Topics

Add Topic: + Subscribe

Topic
No data is available

Back Next

Validate Connection

Save Data Source

Instructions for this tab (Part 2):

4. Enter your **Basic Auth** username and password (Optional – for brokers that support basic auth only).
5. Any number of topics may be subscribed to. To subscribe to a new topic, enter the name of the topic in the field above the data table and press the **Subscribe** button.
6. When complete, press **Validate Connection**. If the connection info is correct, you will now be able to get access to the Transforms Tab.

Tab 2: Transforms

Configure New Data Source

Connection Info | **Transforms**

(Optional) Alter the outgoing Data Object from the PI Web API
Preview and confirm the outgoing Data Object before saving.

Override Property Names

Enter overrides in the form of:
`{"[Original Property]": "[New Name]"}`
For example: `{"Value": "Pressure01"}`

Click to expand

Data Preview

Not Available

Instructions for this tab:

1. (Optional) Override Property Names
 - a. Click on the object symbol { } to open the object editor.
 - b. Override property names by entering the old property name on the left, and the new property name on the right.
 - i. For example, "timestamp": "Time" will rename the "timestamp" property to "Time" in the outgoing data message.
2. When complete, Save the Data Source and exit the configuration pop-up.

3.5.AVEVA PI Web API (Attributes & Elements)

The AVEVA PI Web API data source subscribes to data from individual PI Asset Framework attributes or PI Asset Framework elements (via Asset Framework templates) via the PI Web API.

Tab 1: Connection Info

Configure New Data Source

Connection Info

Enter Name, Connection Info & Validate to Continue.

Data Source Name

Ex: PI Server 1

PI Web API URL

Please enter in the form https://PISERVER.DOMAIN/piwebapi

Basic Auth Username

username

Basic Auth Password

password

Validate Connection

Data Preview

{ }

Save Data Source

Instructions for this tab:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter the **PI Web API URL**.
3. Enter your **Basic Auth** username and password.
4. When complete, press **Validate Connection**. If the connection info is correct, you will now be able to get access to the *AF Server & Database* tab.

Tab 2: AF Server & Database

Configure New Data Source

Connection Info | **AF Server & Database**

Select Data Acquisition Mode, AF Server and Database
A maximum of 1 Server and 1 Database may be selected per PI Web API data source.

Data Acquisition Mode:
☒ Attributes ☐ Templates

AF Server

Name	Connected	Select
BRANDLPI01	false	☐

Asset Database

Name	Path	Select
No data is available		

Data Preview

```
{  
  Message: "Move to Transforms t  
}
```

Save Data Source

Instructions for this tab:

1. Select a Data Acquisition Mode
 - a. **Attributes:** Users can choose individual attributes and create data subscriptions.
 - b. **Template:** Users can choose Asset Framework templates and sync data from every Element using the template
2. Select the desired **AF Server** by clicking on the check box on the right side of the table under the Select column. All Asset Databases in the selected Server will be populated into the Asset Database table.
3. Select the desired **Asset Database** by clicking on the check box on the right side of the table under the Select column. You will now be able to get access to the Choose Attributes Tab or Templates tab.

Tab 3a: Choose Attributes

Configure New Data Source

Connection Info | AF Server & Database | **Choose Attributes** | Transforms

Subscribe to Value Changes for AF Attributes
By default, a unique data subscription is created for each selected Attribute, even if they are in the same Element. Combine data from multiple Attributes within the same Element using "Reference Attributes" in the Transforms tab.

Asset Framework Hierarchy

- Elements
 - Buildings

Select Attributes

Data Re-Check Rate (seconds):
10

Current Attributes | **Selected Attributes**

Attribute	Element
No data is available	

Data Preview

```
{
  Message: "Move to Transforms t
}
```

Save Data Source

Note: Users can navigate to this tab only if "Attribute" mode was selected.

Instructions for this tab:

1. Select an Asset by clicking on the Asset name in the Asset Framework Hierarchy section.
 - a. After selecting an asset, the Current Attributes table will get populated, if the Asset contains any attributes.
2. Subscribe to the desired Attributes by pressing the blue arrow on the right side of the table in the Current Attributes table.
 - a. You will be able to see the selected attributes in the Selected Attributes table.
 - b. You unselect the attributes by pressing the blue arrow on the right side of the table in the Selected Attributes table.
 - c. You may select multiple attributes and each will be sent as individual messages.
3. Enter a rate for the Data Re-Check Rate into the input box.
 - a. This is the rate at which CloudStream will check for new data from the data subscription to the selected attributes. Note that data updating faster than the re-check rate will still be captured by CloudStream. All changes to the subscribed attributes will be captured regardless of the re-check rate.

Tab 3b: Choose Templates

Edit Data Source

Connection Info AF Server & Database **Choose Template** Transforms

Subscribe to Data from Elements using an AF Template
Data from each individual AF Element using the selected Template will be polled based on the chosen Rate.
CloudStream will send an update if the value for any Attribute in an Element has updated since the previous poll.

Data Re-Check Rate (seconds):

Available Templates Refresh Templates

Template	Database
RemoteAssets	CloudStream Demos >
RemoteAssets-Test	CloudStream Demos >

Selected Template

Template	Database
RemoteAssets	CloudStream Demos

Data Preview

```
{
  Name: "default",
  Humidity: "default",
  Lat: "default",
  Lon: "default",
  Pressure: "default",
  Level: "default",
  timestamp: "default",
  Temperature: "[Renamed from Tem
}
```

Save Data Source

Note: Users can navigate to this tab only if “Element” mode was selected.

Instructions for this tab:

1. Select a Template from the Available Templates table.
 - a. You may only select 1 Template at a time.
2. Enter a rate for the Data Re-Check Rate into the input box.
 - a. This is the rate at which CloudStream will check for new data from each element using the selected Template. When the rate is met, CloudStream will collect data from all attributes within each element and compare the values and timestamps to any previously collected data. If a single timestamp or attribute value within the Element has changed since the last time CloudStream polled for data, a new message will be sent.

Tab 4: Transforms

The transforms tab is used to modify the outgoing data messages captured by CloudStream. The Data Preview viewer is updated automatically upon navigating to this tab.

The screenshot shows the 'Configure New Data Source' dialog with the 'Transforms' tab selected. The dialog has four tabs: 'Connection Info', 'AF Server & Database', 'Choose Attributes', and 'Transforms'. The 'Transforms' tab contains three sections: 'Add or Remove Properties from the Data Object', 'Override Property Names', and '(Attributes Only) Add Attribute References'. Each section has a 'Reset' button and a text input field with an expandable object editor icon. The 'Data Preview' panel on the right shows a JSON object with default values for various properties. A 'Save Data Source' button is at the bottom right.

Configure New Data Source

Connection Info | AF Server & Database | Choose Attributes | **Transforms**

(Optional) Alter the outgoing Data Object from the PI Web API
Preview and confirm the outgoing Data Object before saving.

Add or Remove Properties from the Data Object

You may add additional properties with static values, or remove properties. Properties generated by the PI System have the "default" value.

Reset {"Timestamp": "de"} Click to expand

Override Property Names

Enter overrides in the form of: {"[Original Property]": "[New Name]"}
For example: {"Value": "Pressure01"}

Attributes Only: You can dynamically assign the name of the attribute to a property using @AttributeName. For example: {"Value": "@AttributeName"}

Reset {"Value": "Value"} Click to expand

(Attributes Only) Add Attribute References

Add references to other Attributes in the same Element(s).
CloudStream will search for these Attributes and add them to the outgoing object.
Enter Attribute names separated by commas into the list below.

Reset {"ReferenceList": []} Click to expand

Data Preview

```
{
  Timestamp: "default",
  Value: "default",
  UnitsAbbreviation: "default",
  Good: "default",
  Questionable: "default",
  Substituted: "default",
  Annotated: "default"
}
```

Save Data Source

Instructions for this tab:

1. **(Optional) Add or Remove Properties from the Data Object:**
 - a. Click on the object symbol { } to open the object editor.
 - b. Unwanted properties can be deleted. CloudStream will not send deleted properties.
 - c. Additional, static properties can be added. Enter a new property name and the static value for that property.
2. **(Optional) Override Property Names**
 - a. Click on the object symbol { } to open the object editor.
 - b. Override property names by entering the old property name on the left, and the new property name on the right.
 - i. For example, "timestamp": "Time" will rename the "timestamp" property to "Time" in the outgoing data message.
 - ii. Entering the value of "@AttributeName" on the right will dynamically rename the property "Value" to the name of the attribute from PI.
3. **(Optional) Add Attribute References**
 - a. Click on the object symbol { } to open the object editor.
 - b. Add references to other Attributes in the same Element.

- i. Whenever a change of value is received for the subscribed Attribute, CloudStream will search for the value of the reference attribute at the same timestamp as the updated Attribute value, within the same Element of the subscribed Attribute.
- ii. You may enter multiple reference attributes for a single subscribed Attribute.
- iii. If the reference is not found within the AF Element, the reference will be ignored.

When complete, **Save** the Data Source and exit the configuration pop-up.

3.6.AVEVA PI Web API (Tags)

The AVEVA PI Web API (Tags) data source subscribes to data updates from PI Tags.

Tab 1: Connection Info

Configure New Data Source

Connection Info

Enter Name, Connection Info & Validate to Continue.

Data Source Name

Ex: PI Server 1

PI Web API URL

Please enter in the form https://PISERVER.DOMAIN/piwebapi

Basic Auth Username

username

Basic Auth Password

password

Validate Connection

Data Preview

```
{
  Timestamp: "default",
  Value: "default",
  UnitsAbbreviation: "default",
  Good: "default",
  Questionable: "default",
  Substituted: "default",
  Annotated: "default"
}
```

Save Data Source

Instructions for this tab:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter the **PI Web API URL**.
3. Enter your **Basic Auth** username and password.
4. When complete, press **Validate Connection**. If the connection info is correct, you will now be able to access the *Data Server* tab.

Tab 2: Data Server

Configure New Data Source

Connection Info

Data Server

Select Data Acquisition Mode, AF Server and Database

A maximum of 1 Server and 1 Database may be selected per PI Web API data source.

Subscribe to Data

Data from each individual Tag will be polled based on the chosen Rate.
CloudStream will send an update if the value for any Tag has updated since the previous poll.

Data Re-Check Rate (seconds):

10

AF Server

Name	IsConnected	
BRANDLPI01	false	▶
ESD-PIserver-01	false	▶

Data Preview

```
{  
  Timestamp: "default",  
  Value: "default",  
  UnitsAbbreviation: "default",  
  Good: "default",  
  Questionable: "default",  
  Substituted: "default",  
  Annotated: "default"  
}
```

Save Data Source

Instructions for this tab:

1. Enter a rate for the **Data Re-Check Rate** into the input box.
 - a. This is the rate at which CloudStream will check for new data from the data subscription to the selected tags. Note that data updating faster than the re-check rate will still be captured by CloudStream. All changes to the subscribed tags will be captured regardless of the re-check rate.
2. Select the desired PI Data Archive by pressing the blue arrow on the right side of the table. You will be moved to the next tab (Tags tab).

Tab 3: Tags

Configure New Data Source

Connection InfoData ServerTags

Select PI Tags to Poll
Enter a valid filter below to see a list of tags by name.

tagFind Tags

ResultsAdd all

Name	Description
composition.DynamicTag_ThatDied	
composition.StaticTag	
fast_tag	
test1.DynamicTag_ThatDied	
test1.StaticTag	

Back12345...51Next

SelectedClear

Name	Description
No data is available	

BackNext

Data Preview

```
{
  Timestamp: "default",
  Value: "default",
  UnitsAbbreviation: "default",
  Good: "default",
  Questionable: "default",
  Substituted: "default",
  Annotated: "default"
}
```

Save Data Source

Instructions for this tab:

1. Enter a tags name into the top input box and select the Find Tags button.
2. Standard PI Tag masks are supported by the search.
3. All the results will populate the Results table.
4. Select the desired Tags by pressing the blue arrow on the right side of the Results table.
5. You will gain access to the Transforms tab, and you will be able to see all the selected tags in the Selected table.
6. You can unselect a tag by clicking the blue cross on the right side of the Selected table.

Tab 4: Transforms

Connection Info

Data Server

Tags

Transforms

(Optional) Alter the outgoing Data Object from the PI Web API

Preview and confirm the outgoing Data Object before saving.

Add or Remove Properties from the Data Object

You may add additional properties with static values, or remove properties. Properties generated by the PI System have the "default" value.

Reset

{ "Timestamp": "default" }

Click to expand

☐ Expose Tag Name as Inbound Property

Override Property Names

Enter overrides in the form of:
{"[Original Property]": "[New Name]"}
For example: {"Value": "Pressure01"}

Reset

{ "Value": "Value" }

Click to expand

You can dynamically assign the name of the tag to a property using @TagName. For example: {"Value": "@TagName"}

Data Preview

```
{
  Timestamp: "default",
  Value: "default",
  UnitsAbbreviation: "default",
  Good: "default",
  Questionable: "default",
  Substituted: "default",
  Annotated: "default"
}
```

Save Data Source

Instructions for this tab:

1. **(Optional)** Add or Remove Properties from the Data Object:
 - d. Click on the object symbol { } to open the object editor.
 - e. Unwanted properties can be deleted. CloudStream will not send deleted properties.
 - f. Additional, static properties can be added. Enter a new property name and the static value for that property.
 - g. **(Optional)** Enabling "Expose Tag Name as Inbound Property" adds the tag name to the outgoing data object.
2. **(Optional)** Override Property Names
 - c. Click on the object symbol { } to open the object editor.
 - d. Override property names by entering the old property name on the left, and the new property name on the right.
 - iii. For example, "timestamp": "Time" will rename the "timestamp" property to "Time" in the outgoing data message.
 - iv. Entering the value of "@TagName" on the right will dynamically rename the property "Value" to the name of the Tag from PI.

When complete, **Save** the Data Source and exit the configuration pop-up.

CloudStream User Guide

44

3.7.AVEVA PI Web API (Event Frames)

The AVEVA PI Web API (Event Frames) data source checks for and receives new Event Frames.

Tab 1: Connection Info

Configure New Data Source

Connection Info

Enter Name, Connection Info & Validate to Continue.

Data Source Name

Event Frames 1

PI Web API URL

https://example.domain/piwebapi

Basic Auth Username

example.user

Basic Auth Password

.....

Validate Connection

Data Preview

```
{
  Id: "default",
  Name: "default",
  Description: "default",
  StartTime: "default",
  EndTime: "default",
  Severity: "default",
  AcknowledgedBy: "default",
  AcknowledgedDate: "default",
  CanBeAcknowledged: "default",
  IsAcknowledged: "default"
}
```

Save Data Source

Instructions for this tab:

1. Enter a name for the CloudStream Data Source into the top input box.
2. Enter the PI Web API URL.
3. Enter your Basic Auth username and password.
4. When complete, press **Validate Connection**. If the connection info is correct, you will now be able to access the *Asset Server* tab.

Tab 2: Asset Server

On the Asset Server tab, you select a PI Asset Framework Server and can also choose a Data Re-Check Rate.

The screenshot shows the 'Configure New Data Source' dialog box with the 'Asset Server' tab selected. The 'Connection Info' tab is also visible. The 'Select Asset Server' section has a note: 'A maximum of one Asset Server may be selected per PI Web API data source.' Below this, a note states: 'CloudStream will check for new event streams at the below Data Re-Check Rate.' The 'Data Re-Check Rate (seconds):' is set to 10. The 'AF Server' table lists one server: BRANDLPI01, which is not connected. A 'Data Preview' section on the right shows a JSON object with default values. A 'Save Data Source' button is at the bottom right.

Name	Connected	Select
BRANDLPI01	false	

```
{
  Id: "default",
  Name: "default",
  Description: "default",
  StartTime: "default",
  EndTime: "default",
  Severity: "default",
  AcknowledgedBy: "default",
  AcknowledgedDate: "default",
  CanBeAcknowledged: "default",
  IsAcknowledged: "default"
}
```

Save Data Source

Instructions for this tab:

1. Enter a **Data Re-Check** (Default: 10). CloudStream will check for new Event Frames at this rate.
2. Choose an AF Server by pressing the **Select** button to the right of the desired server. CloudStream will only send data for Event Frames in this server. Upon selection of an AF Server, you will be moved to the next tab.

Tab 3: Database

On the Database tab, you choose how CloudStream handles open and closed Event Frames and select an Asset Database.

Edit Data Source

Connection Info | **Asset Server** | **Database** | Filters (Optional) | Transforms (Optional)

Select Database and Event Frame Subscription Mode
A maximum of one Database may be selected per PI Web API data source.

CloudStream will check for new event streams at the below Data Re-Check Rate.

Data Re-Check Rate (seconds):
3

CloudStream will only send events that match the criteria below.

☐ Send an update when a new event frame is created
☐ Send an update when an event frame is closed
☒ Send an update when an event frame is created and when an event frame is closed

Asset Database

Name	Path	Select
ACE	\\BRANDLPI01\ACE	▶
ACE-REDUCED	\\BRANDLPI01\ACE-REDUCED	▶
AnalyticsTestDB	\\BRANDLPI01\AnalyticsTestDB	▶
CloudStream Demos	\\BRANDLPI01\CloudStream Demos	▶
CloudStream Dev	\\BRANDLPI01\CloudStream Dev	▶
Configuration	\\BRANDLPI01\Configuration	▶
Database16	\\BRANDLPI01\Database16	▶

Data Preview

```
{
  Id: "default",
  Name: "default",
  Description: "default",
  StartTime: "default",
  EndTime: "default",
  Severity: "default",
  AcknowledgedBy: "default",
  AcknowledgedDate: "default",
  CanBeAcknowledged: "default",
  IsAcknowledged: "default",
  Pressure: "default"
}
```

Save Data Source

Instructions for this tab:

- Enter a **Data Re-Check** (Default: 10). CloudStream will check for new Event Frames at this rate.
- Choose how CloudStream handles open and closed Event Frames:
 - Send an update when a new event frame is created:** Use this option if you want CloudStream to only include Event Frames that have a StartTime later than the time of the last poll.
 - Send an update when an event frame is closed:** Use this option if you want CloudStream to only include Event Frames that have an EndTime later than the time of the last poll.
 - Send an update when an event frame is created and when an event frame is closed:** Use this option if you want CloudStream to include any Event Frames that have a StartTime or an EndTime later than the time of the last poll.
- Select an Asset Database by pressing the **Select** button next to the desired Asset Database. CloudStream will search this Database for new Event Frames that meet the criteria specified in the configuration. Upon selecting an Asset Database, you will be moved to the Filters (Optional) tab.

Tab 4: Filters (Optional)

On the Filters tab, you are given the option to specify the configuration of five filters: **Name**, **Severity**, **Category**, **Template**, and **Primary Element**

The screenshot shows the 'Configure New Data Source' dialog with the 'Filters (Optional)' tab selected. The dialog has a title bar with a close button. Below the title bar are five tabs: 'Connection Info', 'Asset Server', 'Database', 'Filters (Optional)', and 'Transforms (Optional)'. The 'Filters (Optional)' tab is active and contains the following sections:

- (Optional) Define Event Frame Filters**: A header section with a note: 'CloudStream will check each incoming Data Frame for the following characteristics:'.
- Event Frame Name**: A text input field with a placeholder 'Enter a search filter (ex. "**Engine Over Temp**")'.
- Severity**: A dropdown menu with 'Any' selected.
- Category**: A text input field with a placeholder 'Search for a category', a 'Q Search' button, and an 'X Remove' button.
- Template**: A text input field with a placeholder 'Search for a template', a 'Q Search' button, and an 'X Remove' button.
- Primary Element Name**: A text input field with a placeholder 'Enter a search filter (ex. "**Asset1**")'.

On the right side of the dialog is a **Data Preview** section showing a JSON object:

```
{
  Id: "default",
  Name: "default",
  Description: "default",
  StartTime: "default",
  EndTime: "default",
  Severity: "default",
  AcknowledgedBy: "default",
  AcknowledgedDate: "default",
  CanBeAcknowledged: "default",
  IsAcknowledged: "default"
}
```

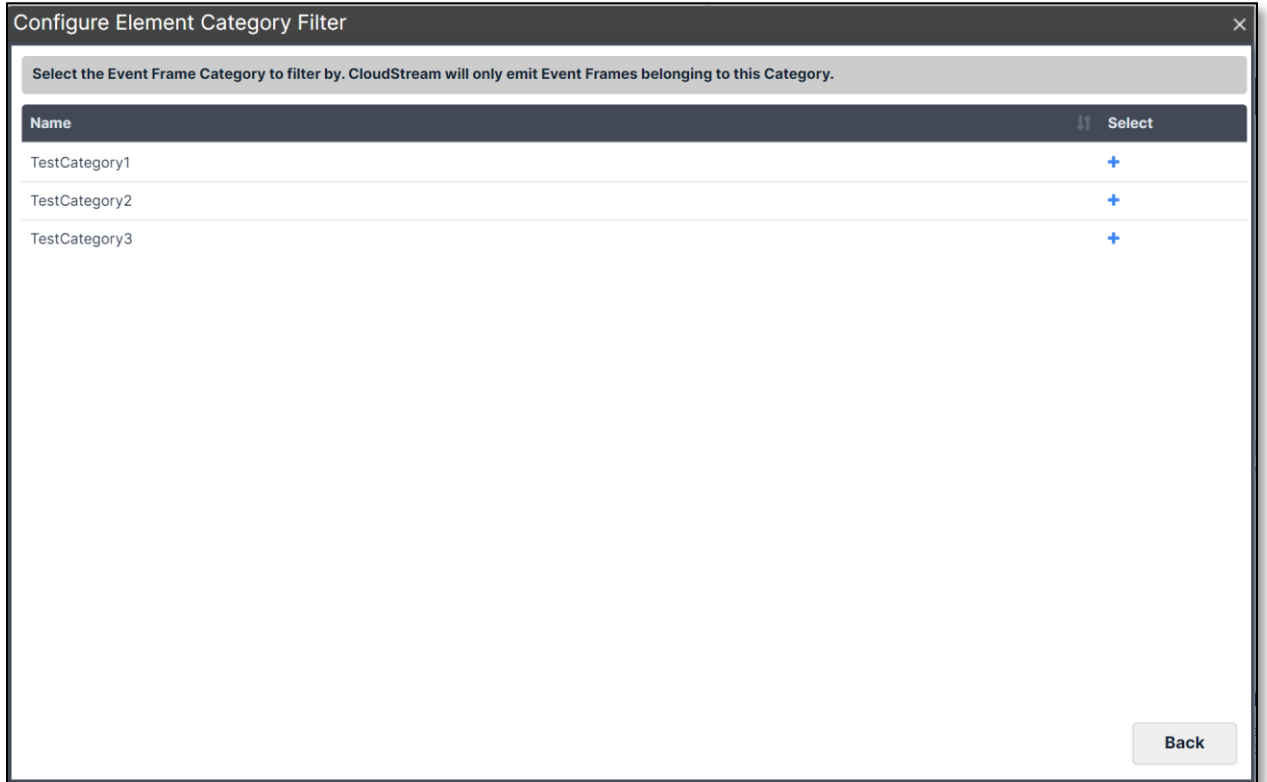
At the bottom right of the dialog is a 'Save Data Source' button.

Instructions for this tab (Part 1):

1. (Optional) Provide a filter that will be applied to the **Name** property of incoming Event Frames. The filter uses the same syntax as AVEVA's Search Query: <https://docs.aveva.com/bundle/pi-web-api-reference/page/help/topics/search-query-syntax.html>
2. (Optional) Select a **Severity** that all Event Frames will need to have to be included.

Instructions for this tab (Part 2):

- (Optional) Select an **Element Category** that all Event Frames will need to belong to in order to be included. This is done by pressing the **Search** button, bringing up the Element Category Selection Popup:



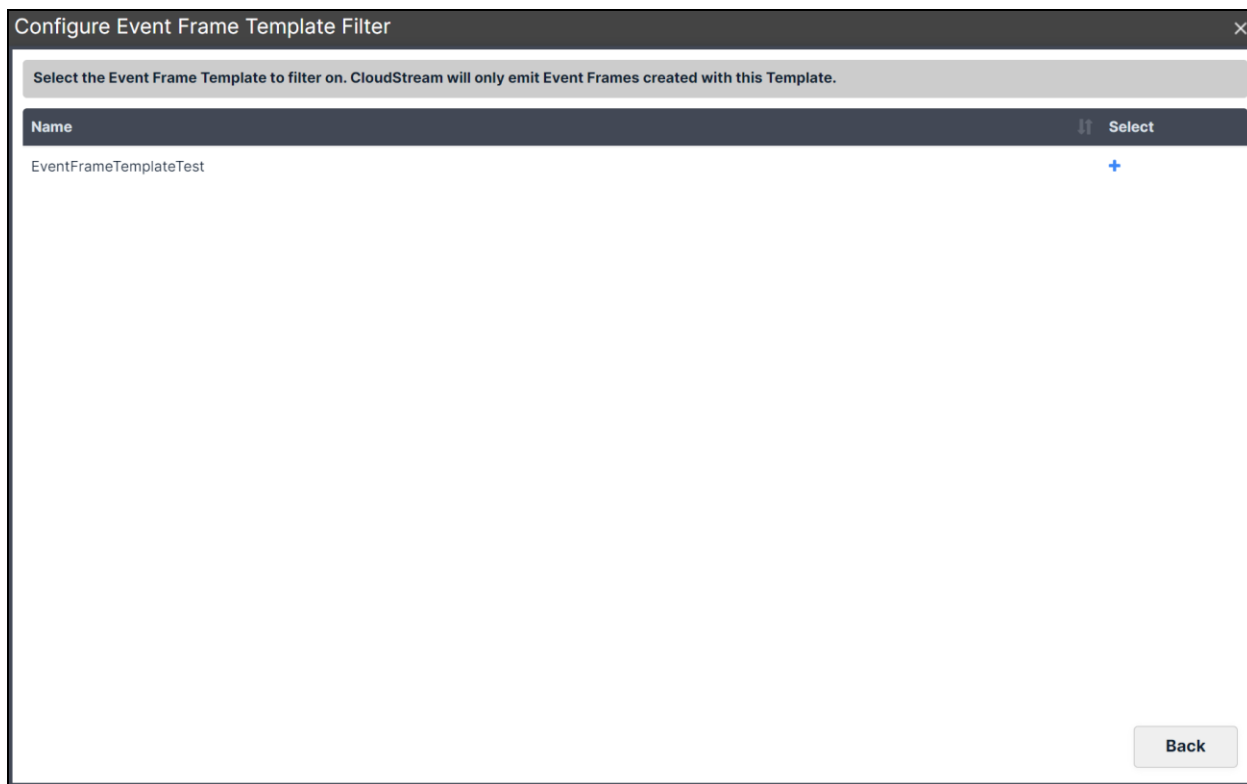
The dialog box is titled "Configure Element Category Filter" and has a close button (X) in the top right corner. Below the title bar is a grey instruction bar that reads: "Select the Event Frame Category to filter by. CloudStream will only emit Event Frames belonging to this Category." Below this is a table with three rows. The first row is a header with "Name" on the left and "Select" on the right, separated by a double vertical bar icon. The second row contains "TestCategory1" and a blue plus icon. The third row contains "TestCategory2" and a blue plus icon. The fourth row contains "TestCategory3" and a blue plus icon. At the bottom right of the dialog is a "Back" button.

Name	Select
TestCategory1	+
TestCategory2	+
TestCategory3	+

Back

Instructions for this tab (Part 3):

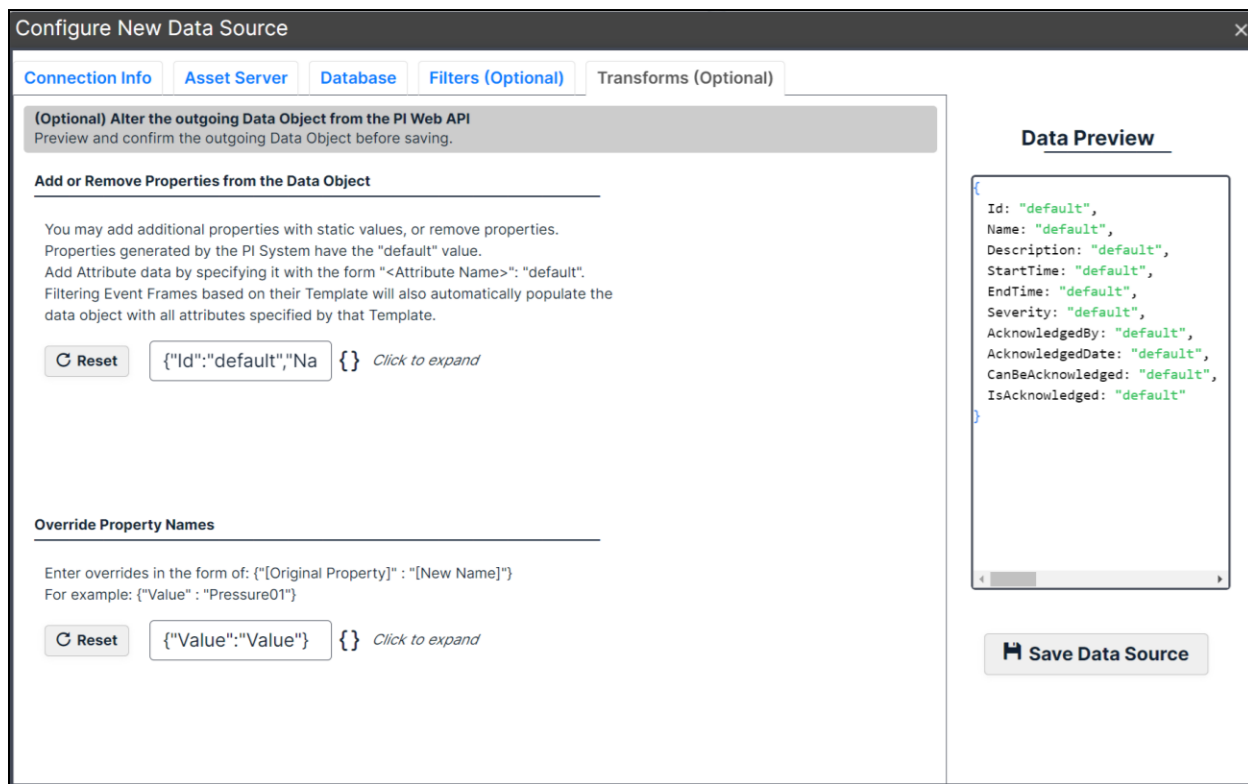
- (Optional) Select a **Template** that all Event Frames must have been created with in order to be included. This is done by pressing the **Search** button, bringing up the Template Selection Popup:



After selecting a Template, the Data Object Preview will be populated with the Attributes associated with that Template

- (Optional) Provide a filter that will be applied to the Name property of the first Referenced Element of all incoming Event Frames. The first Referenced Element of each Event Frame must match the filter in order to be included. The filter uses the same syntax as AVEVA's Search Query: <https://docs.aveva.com/bundle/pi-web-api-reference/page/help/topics/search-query-syntax.html>

Tab 5: Transforms (Optional)



Configure New Data Source

Connection Info | Asset Server | Database | Filters (Optional) | **Transforms (Optional)**

(Optional) Alter the outgoing Data Object from the PI Web API
Preview and confirm the outgoing Data Object before saving.

Add or Remove Properties from the Data Object

You may add additional properties with static values, or remove properties. Properties generated by the PI System have the "default" value. Add Attribute data by specifying it with the form "<Attribute Name>": "default". Filtering Event Frames based on their Template will also automatically populate the data object with all attributes specified by that Template.

{"Id":"default","Na"

Override Property Names

Enter overrides in the form of: {"[Original Property]" : "[New Name]"}. For example: {"Value" : "Pressure01"}

{"Value":"Value"}

Data Preview

```
{
  Id: "default",
  Name: "default",
  Description: "default",
  StartTime: "default",
  EndTime: "default",
  Severity: "default",
  AcknowledgedBy: "default",
  AcknowledgedDate: "default",
  CanBeAcknowledged: "default",
  IsAcknowledged: "default"
}
```

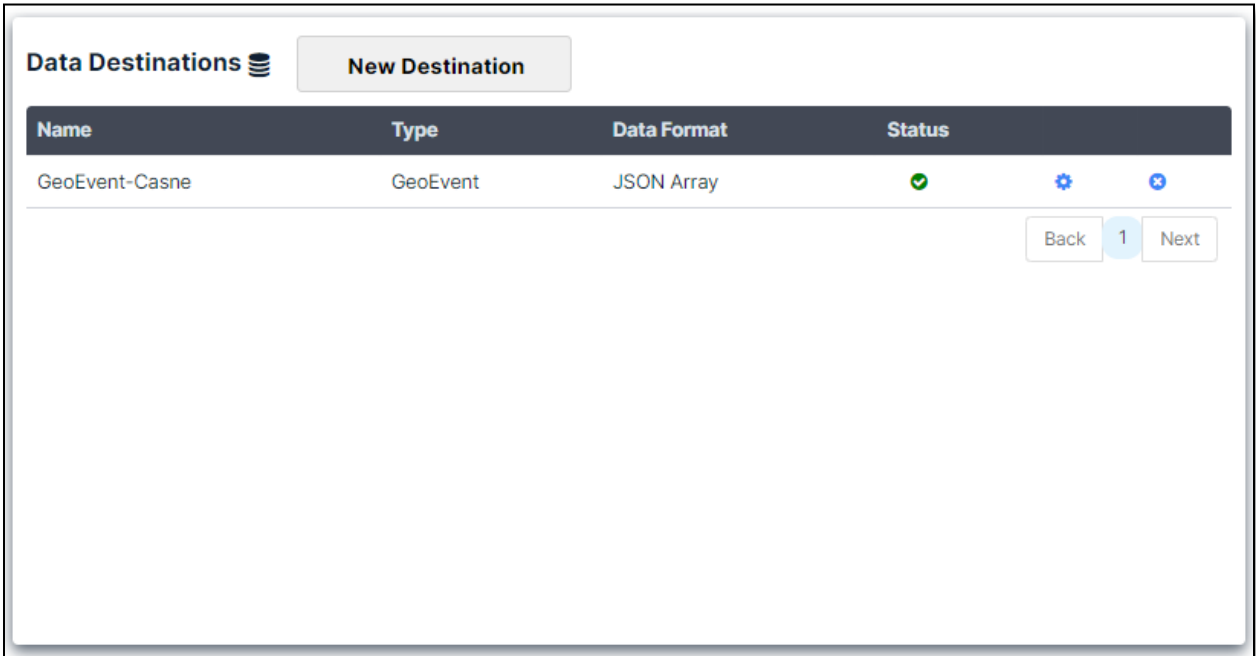
Instructions for this tab:

1. **(Optional) Add or Remove Properties from the Data Object:**
 - a. Click on the object symbol { } to open the object editor.
 - b. Unwanted properties can be deleted. CloudStream will not send deleted properties.
 - c. Additional static properties can be added. Enter a new property name and the static value for that property.
 - d. To include data from additional **Attributes**, and a line of the form "<Attribute Name>": "default" to the **Data Object**.
2. **(Optional) Override Property Names**
 - a. Click on the object symbol { } to open the object editor.
 - b. Override property names by entering the old property name on the left, and the new property name on the right.
 - i. For example, "timestamp": "Time" will rename the "timestamp" property to "Time" in the outgoing data message.

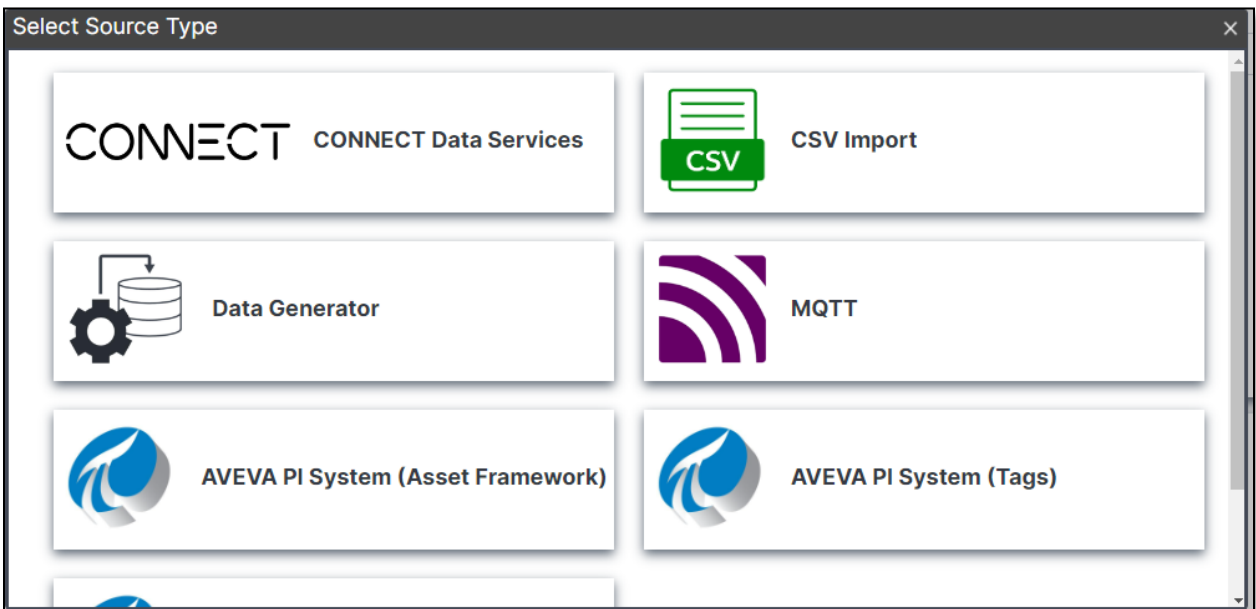
When complete, **Save** the Data Source and exit the configuration pop-up.

4. CloudStream Destination Creation

To create a new CloudStream Destination, click on the “**New Destination**” button from the Config page. Existing destinations may be edited by pressing the gear button. Destinations may be deleted by pressing the delete button (Note: a Destination cannot be deleted if it is part of a Service).



This will open the Destination Selection pop-up:



Click on any destination to open up the corresponding Destination Configuration window.

4.1. AVEVA CONNECT Data Services

The AVEVA CONNECT Data Services Destination sends data to AVEVA CONNECT Data Services SDS Streams.

Tab 1: Connection Info

The screenshot shows a window titled "Configure New Data Destination" with a close button (X) in the top right corner. The window is divided into two main sections. The left section, titled "Connection Info", contains a grey bar with the text "Enter Name, Connection Info & Validate to Continue." Below this are four input fields: "CloudStream Destination Name" (with an example "Ex: CONNECT Data Services 1"), "Client ID" (with placeholder text "Please enter your CONNECT Data Services Client ID"), "Client Secret" (with placeholder text "Please enter your CONNECT Data Services Client Secret"), and a "Select Region:" dropdown menu (currently showing "Select Region"). At the bottom of this section is a checkbox labeled "Execute Destination Processing at Controller? (WSS for Edge-to-Cloud data transport)". A "Validate Connection" button is at the bottom left. The right section, titled "Data Preview", shows a large empty box with curly braces "{ }" at the top. A "Save Data Source" button is at the bottom right.

Instructions for this tab:

1. Enter the name for the Data Destination into the top input box.
2. Enter your AVEVA CONNECT Data Services **Client ID** and **Client Secret**.
3. Select your Data Hub Region (Currently supported: **US-West**)
4. Choose if Data should be routed from the Node or Controller
 - a. Select this option if you need additional security for sending messages to Data Hub. This will use Secure Web Socket security to first route the message to the controller.
5. Click "Validate Connection".
 - a. If successful, you will automatically move to the next configuration step ("**Namespace**" Tab).
 - b. In unsuccessful, you will be given an error message. Please correct the **Client ID**, **Client Secret**, or **Region**.
 - c. A Data Source may be saved at any time once Validation is complete.

Tab 2: Namespace

Configure New Data Destination

Connection Info

Namespace

Enter Tenant ID and Select Namespace

Enter Tenant ID

Get Namespaces

☒ Streams

☐ Events

Available Namespaces

Name	Region	Description	Select
No data is available			

Data Preview

Save Data Source

Instructions for this tab:

1. Enter your Tenant ID and click “**Get Namespaces**”.
2. Select AVEVA CONNECT Streams or AVEVA CONNECT Events
3. Select the Namespace containing the Streams you want to send data to.
 - a. Only 1 Namespace can be selected per CloudStream AVEVA CONNECT Data Services Destination
 - b. Once you select a **Namespace** you will automatically navigate to the “**Streams**” tab or “**Events**” tab, based on which was previously selected.

Tab 3: Streams

Configure connection to AVEVA Data Hub

Connection Info | Namespace | Streams

Select Streams to receive data from CloudStream

Available Streams

Name	TypeId	Select
Test1	TEST	▶

Selected Streams

Name	TypeId	Remove
No data is available		

Data Preview

```
{ }
```

Save Data Source

Instructions for this tab:

1. Select Streams you want to send data to by pressing the button under the “**Select**” tab.
 - a. You may select multiple streams, but they must have the same **TypeId**.
 - b. An error will be generated if Streams with different **Typelds** are selected.
 - c. Upon selecting a Stream, the Data Preview will automatically update with the selected **TypeId**.
2. You may de-select a stream by pressing the button under the “**Remove**” column.
3. When complete, **Save** the Data Destination and exit the configuration pop-up.

Tab 4: Events

Configure New Data Destination

Connection Info

Namespace

Events

Select Events to receive data from CloudStream

Available Event Types

Name	TypeId	Select
BaseEvent	BaseEvent	
SimpleEvent	c6389280-5744-4d45-b354-2471cf444afb	
TestEvent	23ea4f68-a11f-4a0f-a20e-0a0dfd290650	
DemoEvent	87b8db1e-f820-40df-bb8d-77a98f8d12a3	
AssetEvent	d9221f5a-d9a4-4f54-9266-37a1d432b178	
NewEvent	5b38f8ba-ba00-4551-8881-8440f4e2a2d0	

New Event Type

Selected Event Types

Name	TypeId	Remove
No data is available		

Data Preview

```
{
}
```

Save Data Source

Instructions for this tab:

- Select Event Type you want to send data to by pressing the button under the “**Select**” tab.
 - You may only select one Event Type.
 - An error will be generated if more then one Event Type are selected.
 - Upon selecting a Event Type, the Data Preview will automatically update with the selected **TypeId**.
- (Optional) Select “New Event Type” to create a new Event Type.
 - Upon selecting you will automatically navigate to the “Create Event” tab.
- You may de-select a Event Type by pressing the button under the “**Remove**” column.
- When complete, **Save** the Data Destination and exit the configuration pop-up.

Tab 5: Create Event

Configure New Data Destination

Connection InfoNamespaceEventsCreate Event

Create Event Type to receive data from CloudStream

New Event Type Name

Ex: SampleEvent

Authorization Tag

BaseAuthorizationTag

Create Authorization Tag

Event Type Attributes

Add Attribute: String + Add

Attribute	Type	
name	String	×
id	String	×
modifiedDate	DateTime	×
createdDate	DateTime	×
createdByUser	String	×
authorizationTags	String	×

Create Event Type

Data Preview

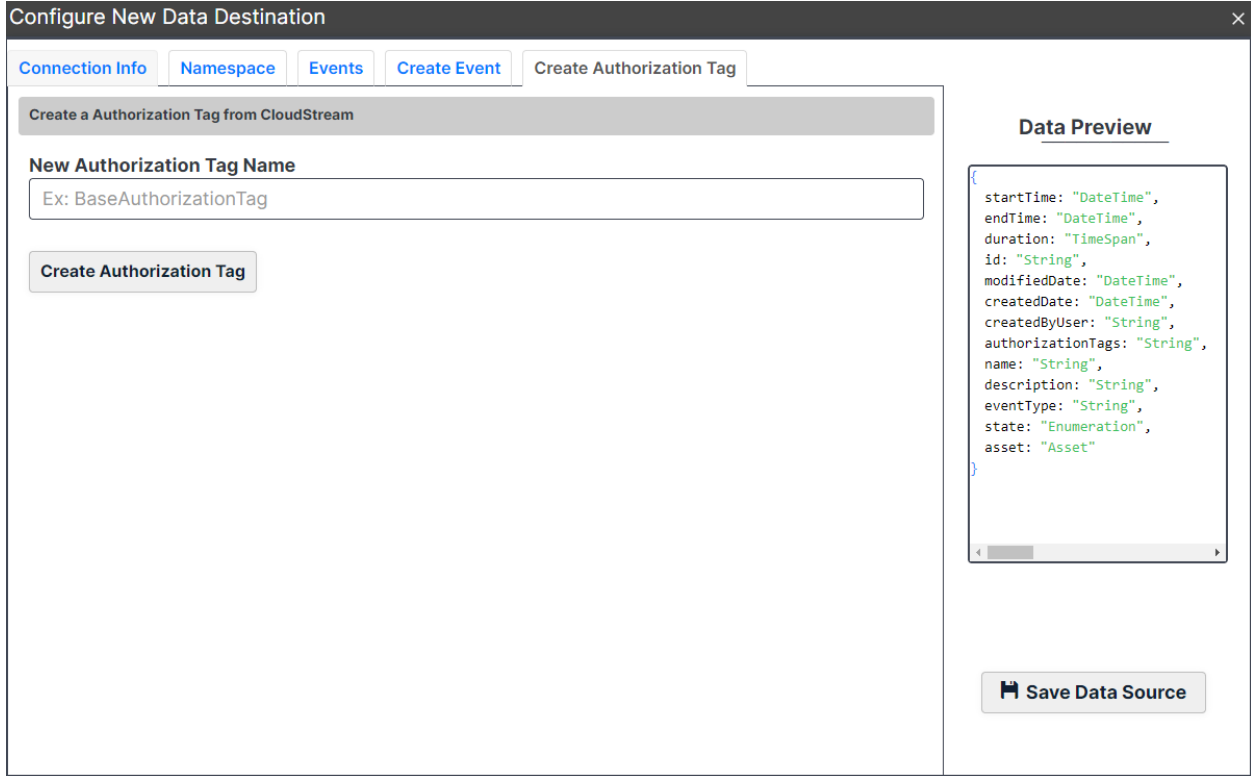
```
{  startTime: "DateTime",  endTime: "DateTime",  duration: "TimeSpan",  id: "String",  modifiedDate: "DateTime",  createdDate: "DateTime",  createdByUser: "String",  authorizationTags: "String",  name: "String",  description: "String",  eventType: "String",  state: "Enumeration",  asset: "Asset"}
```

Save Data Source

Instructions for this tab:

1. Enter the name for the New Event Type Name into the top input box
2. Select a Authorization Tag (default Authorization Tag is "BaseAuthorizationTag").
3. (Optional) Select "Create Authorization Tag" to create a new Authorization Tag.
 - a. Upon selecting you will automatically navigate to the "Create Authorization Tag" tab.
4. Any number of attributes may be added. To add a new attribute, enter the name of the attribute in the field above the data table then select the type for the attribute and press the **Add** button.
 - a. The data table contains default Attributes.
 - b. Attributes can be removed by clicking the blue cross on the right side of the table.

Tab 6: Create Authorization Tag



The screenshot shows a dialog box titled "Configure New Data Destination" with a close button (X) in the top right corner. It features four tabs: "Connection Info", "Namespace", "Events", and "Create Authorization Tag". The "Create Authorization Tag" tab is active. Below the tabs, there is a header "Create a Authorization Tag from CloudStream". A section titled "New Authorization Tag Name" contains a text input field with the placeholder text "Ex: BaseAuthorizationTag". Below this input field is a button labeled "Create Authorization Tag". To the right of the main form area is a "Data Preview" section containing a JSON schema definition for an authorization tag. At the bottom right of the dialog is a button labeled "Save Data Source".

Configure New Data Destination

Connection Info Namespace Events Create Event Create Authorization Tag

Create a Authorization Tag from CloudStream

New Authorization Tag Name

Ex: BaseAuthorizationTag

Create Authorization Tag

Data Preview

```
{
  startTime: "DateTime",
  endTime: "DateTime",
  duration: "TimeSpan",
  id: "String",
  modifiedDate: "DateTime",
  createDate: "DateTime",
  createdByUser: "String",
  authorizationTags: "String",
  name: "String",
  description: "String",
  eventType: "String",
  state: "Enumeration",
  asset: "Asset"
}
```

Save Data Source

Instructions for this tab:

1. Enter the name for the New **Authorization Tag** into the top input box. Then press the **Add** button

4.2. Azure Event Hub

The Azure Event Hub Destination sends data to resources in an Azure Event Hub namespace.

Configure New Data Destination

Enter Destination Name & Policy Info

CloudStream DestinationName

Ex: Azure Event Hubs 1

Azure Instance Information

Event Hubs Namespace

Please enter the name of your Event Hub Namespace

Resource Name

<instance_name>, <queue_name>, or <topic_name>/Subscription/<subscription_name>

Authentication

SAS Policy Name

Please enter your SAS Policy name

SAS Policy Primary Key

Please enter your SAS Policy primary key

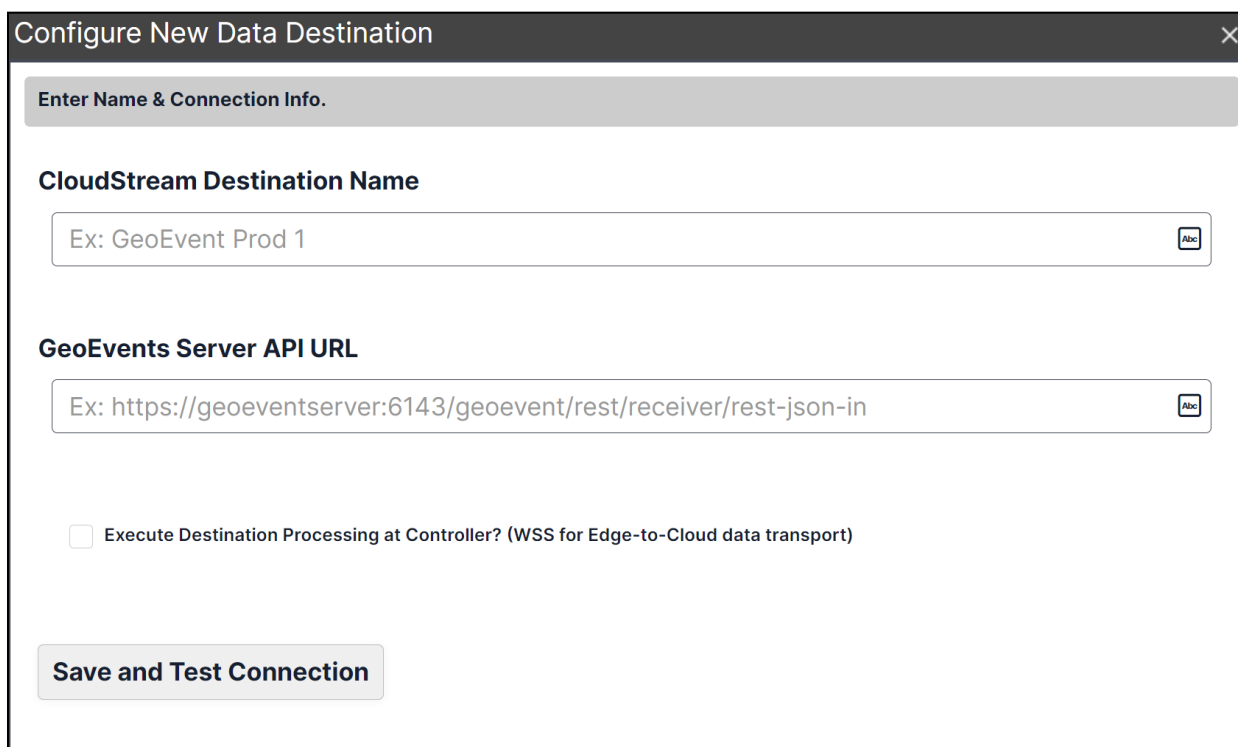
Save

Instructions for this Destination:

1. Enter the name for the Data Destination into the top input box.
2. Under **Azure Instance Information**, enter your **Azure Event Hubs** Namespace and the resource that data should be published to.
3. Under **Authentication**, enter the name of a valid **Shared Access Policy** and the primary key associated with that policy
4. When complete, **Save** the Data Destination and exit the configuration pop-up.

4.3. ESRI GeoEvent Server

The ESRI GeoEvent Server Destination sends data to an ESRI GeoEvent Server.



The screenshot shows a 'Configure New Data Destination' dialog box with a close button (X) in the top right corner. The dialog has a title bar and a main content area. At the top, there is a grey header bar with the text 'Enter Name & Connection Info.'. Below this, the 'CloudStream Destination Name' section has a text input field with the placeholder text 'Ex: GeoEvent Prod 1' and a small 'ABC' icon button to its right. The 'GeoEvents Server API URL' section has a text input field with the placeholder text 'Ex: https://geoeventserver:6143/geoevent/rest/receiver/rest-json-in' and a small 'ABC' icon button to its right. Below these fields is a checkbox labeled 'Execute Destination Processing at Controller? (WSS for Edge-to-Cloud data transport)'. At the bottom left of the dialog is a 'Save and Test Connection' button.

Instructions for this Destination:

1. Enter the name for the Data Destination into the top input box.
2. Enter the URL for the desired **GeoEvents Server API** into the next input box.
3. If desired, enable processing at the controller by ticking the checkbox.
4. When complete, **Save** the Data Destination and exit the configuration pop-up.

4.4. MQTT

The MQTT destination connects to a MQTT broker and publishes messages to any number of MQTT topics.

Tab 1: Connection Info

Configure New Data Destination

Connection Info

Enter Name, Connection Info & Validate to Continue.

CloudStream Destination Name

Ex: MQTT-out

Connection

Broker URL

Ex: localhost, mqtt.dev

Broker Port

Ex: 1883

Broker Encryption

☒ Unencrypted (tcp)

☐ Encrypted (tcps)

QoS

QoS 0

Data Preview

Not Available

Instructions for this tab (Part 1):

5. Enter the name for the CloudStream Data Source into the top input box.
6. Enter the server name and port for the MQTT broker.
7. Choose encryption mode for the broker (TCP or TCPS)
8. Select a QoS (default QoS is 0)
 - QoS 0 (At most once) provides a best-effort delivery system where the sender does not wait for an acknowledgment or guarantee delivery. The recipient does not confirm receipt, and the sender neither stores nor retries sending the message.
 - QoS 1 (At least once) ensures that the message is delivered to the receiver at least once. When a message is published with QoS 1, the sender retains a copy of the message until it receives a PUBACK packet from the receiver, confirming successful delivery.
 - QoS 2 (Exactly once) offers the highest level of service in MQTT, ensuring that each message is delivered exactly once. This is achieved through a four-step handshake process between the sender and the receiver.

Basic Auth Username (Optional)

username

Basic Auth Password (Optional)

.....

MQTT Topics

Add Topic: + Subscribe

Topic
No data is available

Back Next

Validate Connection

Save Data Source

Instructions for this tab (Part 2):

7. Enter your **Basic Auth** username and password (Optional – for brokers that support basic auth only).
8. Any number of topics may be subscribed to. To subscribe to a new topic, enter the name of the topic in the field above the data table and press the **Subscribe** button.

When complete, press **Validate Connection**. If the connection info is correct, you will now be able to save the destination.

4.5. AVEVA PI Web API (Tags)

The PI Web API (Tags) destination publishes data directly to PI Tags. This data contains considerable data mapping to ensure that the PI Data Archive can receive the incoming data.

Tab 1: Connection Info

Configure New Data Destination [X]

Connection Info

Enter Name, Connection Info & Validate to Continue.

Data Source Name

Ex: PI Server 1

PI Web API URL

Please enter in the form https://PISERVER.DOMAIN/piwebapi

Basic Auth Username

username [Eye Icon]

Basic Auth Password

password [Eye Icon]

Validate Connection

Instructions for this tab:

1. Enter the name for the CloudStream Data Destination into the top input box.
2. Enter the **PI Web API URL**.
3. Enter your **Basic Auth** username and password.
4. When complete, press **Validate Connection**. If the connection info is correct, you will now be able to save the Data Destination.

Tab 2: Data Server

Configure New Data Destination

Connection Info

Data Server

Data Mapping

Select Data Acquisition Mode, AF Server and Database

A maximum of 1 Server and 1 Database may be selected per PI Web API data source.

AF Server

Name	IsConnected	
BRANDLPI01	false	▶
ESD-PIserver-01	false	▶

Instructions for this tab:

1. Select the desired **Data Server** by pressing the blue arrow on the right side of the table. You will be moved to the next tab.

Tab 3: Data Mapping

Configure New Data Destination

Connection Info

Data Server

Data Mapping

Map incoming properties to PI Tag properties

Tag Name:

☒ From JSON Property ☐ From PI System

Name	Description
No data is available	

BackNext

Value:

☒ From JSON Property ☐ Static Value

Units Abbreviation

☒ From JSON Property ☐ Static Unit Abbreviations

Instructions for this tab (Part 1):

1. Under **Tag Name**, either search the **Data Server** for the desired tag by entering a search mask and pressing **Search for Tags** or by entering a JSON property of the incoming message containing the **Tag** name.
 - a. If the Tag name is contained within an incoming JSON property, then the tag name will be dynamic based on the incoming message. Otherwise, data will only be sent to the tag specified from the PI Data Archive.

CloudStream User Guide

65

Configure New Data Destination

Connection Info

Data Server

Data Mapping

Map incoming properties to PI Tag properties

Timestamp:

☒ From JSON Property ☐ Use Send Time

Enter the property name containing the timestamp from the incoming message

Questionable:

☒ From JSON Property ☐ True ☐ False

Enter the property containing the Questionable boolean from the incoming message

Good:

☒ From JSON Property ☐ True ☐ False

Enter the property containing the Good boolean from the incoming message

Save Data Destination

Instructions for this tab (Part 2): Data Mapping

2. For each of the following **PI Tag Properties**, either enter a JSON property of the incoming object containing the respective **PI Tag Property** or enter a specific value:
 - a. **Value**
 - b. **Units Abbreviation**
 - c. **Timestamp**
 - d. **Questionable**
 - e. **Good**

4.6. pSQL

The pSQL Destination sends data to a pSQL database.

Note: This connector uses a pSQL driver subservice that must be initialized alongside the CloudStream node. Please contact your CloudStream installer for additional information.

Tab 1: Connection Info

Configure New Data Destination

Connection Info

Enter Name, Connection Info & Validate to Continue.

CloudStream Destination Name

Ex: pSQL DB 1

Connection

Host URL

Ex: http://sqlhost.com

Database Port

Ex: 5432

Database Name

Ex: SQL_DB

Authentication

Username

Password

Show Token

Validate Connection

Data Table Preview

Save Data Source

Instructions for this tab:

1. Enter the name for the CloudStream Data Destination into the top input box.
2. Under **Connection**, enter your database host's URL and port, as well as the name of the database to insert into.
3. Under **Authentication**, enter a username and password for a user with sufficient permission.
4. When complete, press **Validate Connection**. If connection info is correct, you will be moved to the next tab.

Note: The "Show Token" button generates an authentication token for the pSQL driver subservice to securely connect to the CloudStream Node. This token is only used by CloudStream installers during subservice configuration.

Tab 2: Tables

Configure New Data Destination

Connection Info

Tables

Select a table to insert into

Selected Table

test

Available Tables

Table Name	
test	▶

Data Table Preview

```
{  
  id: "integer"  
}
```

Save Data Source

Instructions for this tab:

1. Select a table from the list of tables. The columns of the selected table and their respective types are now visible in the **Data Table Preview**. Each pSQL Destination can only have one table.
2. When complete, **Save** the Data Destination and exit the configuration pop-up.

4.7. Vantiq

The Vantiq destination publishes JSON object data directly to a Vantiq topic.

Configure New Data Destination

Destination Name

Ex: Vantiq Prod 1

Vantiq URL

Ex: https://dev.vantiq.com

Vantiq Topic

Ex: /MyTopic/Data

Access Token

token

☐ Execute Destination Processing at Controller? (WSS for Edge-to-Cloud data transport)

Save and Test Connection

Instructions for this Destination:

1. Enter the name for the Data Destination into the top input box.
2. Enter the URL where **Vantiq** is located.
3. Enter the access token corresponding to the appropriate **Vantiq** namespace.
4. If desired, enable processing at the controller by ticking the checkbox.
 - a. This causes the data to first be routed to the controller before submitting to the remote Vantiq instance. This ensures that data is kept within WSS encryption and may be useful for edge nodes without certificates.
5. When complete, **Save** the Data Destination and exit the configuration pop-up.

5. Data Injectors

Data Injectors are a way to modify your data as it streams through Casne CloudStream. Injectors are python scripts executing against data messages in real time. The input of a Data Injector is data from a CloudStream source, and the output of a Data Injector is sent to a CloudStream destination. A CloudStream service can use any number of Data Injectors, and each Injector is executed in the order indicated by the user. The same Data Injector may be used multiple times by the same CloudStream Service. However, each evocation of a Data Injector (and its respective variables) is unique per CloudStream Service.

Caution: Injectors are executed against streaming, in-flight data messages. Therefore, consideration must be taken towards reducing processing times. If the script execution time is >2 minutes, the script will be terminated, and the un-modified results will be sent to the CloudStream Destination. If your processing time is consistently greater than the rate at which data is received from the CloudStream Source, an unmanageable backlog of data may accumulate, leading to an eventual system crash. This will be visible as a significant increase in CPU and RAM usage and warnings written to the CloudStream logs.

Data Injectors are available in both Edge and Cloud installations of Casne CloudStream, however the architecture differs between the two.

Edge Installations:

Data Injectors are processed in a dedicated docker container or windows application running in parallel with Casne CloudStream. Data is encrypted and transferred locally between the applications using WSS (secure web socket). This service may be started or stopped at any time by the customer and is installed automatically when CloudStream is installed (if Data Injectors have been licensed by the customer). Each Edge node of CloudStream must have a dedicated Data Injector service.

Managed Cloud Installations:

Data Injectors are processed in a dedicated cloud hosted container (managed by Casne Engineering). Data is encrypted and transferred between CloudStream and the dedicated container using WSS (secure web socket). This container can be initialized, destroyed, or rebooted using the CloudStream control interface for a CloudStream node (see [section 2.6.3](#)). Data Injector containers are unique to the Node/Customer they are connected to.

5.1 Configuring Data Injectors

Data Injectors are configured from the Config page of the Casne CloudStream System Control web application, within the “Injectors” tab.

Begin by pressing “New Injector”:

Tab 1: Script

Configure CloudStream Injector

Script **Variables**

Paste script and upload or select existing script

Data Injector Name

Ex: Transform 1

Script Name

(Leave blank for an autogenerated name)

Injector Code Block

```
#-----CloudStream Injector Example Code-----#
#
# Every Injector script is passed a single parameter at runtime:
#   Name: data
#   Type: JSON Object or Array of JSON Objects (depending on the CloudStream Source)
#   Description: Contains the data returned by the CloudStream source
#
# Your code should modify the contents of the 'data' parameter. When the script completes, the 'data' parameter
# is automatically returned and passed to the CloudStream Destination.
#
# Note: Any other return or print statements are ignored
# Note: 'data' must remain as a JSON Object or Array of JSON for CloudStream to continue processing.
#
# CAUTION: Consideration must be taken towards reducing processing times. If script execution time
#           is >2 minutes, the script will be terminated and the un-modified results will be sent to the
```

Or, upload a .py file:

Upload Python Script

Save and Exit

Instructions for this tab:

1. Enter the name for the CloudStream Data Injector into the top input box. This is the name that will be used to identify it from logs and within the service creator.
2. (Optional) Enter a name for the Injector Script. If multiple injectors use the same script name, they will reference the same uploaded python script.

3. Enter python code into the Injector Code block OR uploaded a .py file.

- a. Uploading a .py file will automatically fill the Injector Code Block with your script text.
- b. Example code is included with every injector. The example code contains importing data processing considerations and warnings, and a copy of this code is included for reference here:

```
#-----CloudStream Injector Example Code-----#
#
# Every Injector script is passed a single parameter at runtime:
#   Name: data
#   Type: JSON Object or Array of JSON Objects (depending on the CloudStream Source)
#   Description: Contains the data returned by the CloudStream source
#
# Your code should modify the contents of the 'data' parameter. When the script completes, the 'data' parameter
# is automatically returned and passed to the CloudStream Destination.
#
#   Note: Any other return or print statements are ignored
#   Note: 'data' must remain as a JSON Object or Array of JSON for CloudStream to continue processing.
#
#   CAUTION: Consideration must be taken towards reducing processing times. If script execution time
#             is >2 minutes, the script will be terminated, and the un-modified results will be sent to the
#             CloudStream Destination. If your processing time is consistently greater than the rate at
#             which data is received from the CloudStream Source, an unmanageable backlog of data
#             may accumulate, leading to an eventual system crash. This will be visible as a significant
#             increase in CPU and RAM usage.
#
# Any Variables initialized in the 'Variables' tab will have their values saved in memory between script invocations.
# You may access these variables and their values in your code without declaring them.
#
# You may import any modules or packages that have been installed.
# You may install or check installed modules/packages at any time from the 'System' page by clicking on
# 'Import Modules'
#
# Example of modifying 'data' as an object:
#   data['parameter'] = 100
#
# Example of iterating 'data' as an array:
#   for i, record in enumerate(data):
#       data[i]['parameter'] = 100
#-----#
```

- c. It is recommended to test your script before running it within a service. Per the example code, CloudStream will always send a data object or array to the injector for processing. Ensure that your code properly compiles and returns a valid JSON object that complies with the CloudStream destination.
 - i. Compilation or runtime errors will be visible from the CloudStream error log while the injector is running. In the case of an injector error, the unmodified data will be sent to the destination.
 - ii. The list of expected data types per CloudStream Source and Destination are indicated in the “Data Format(s)” column. The full list is as follows:

Table of CloudStream Connector Data Formats

Connector Type	Name	Data Format(s)
Source	CSV	Array of JSON
Source	CONNECT Data Services	JSON Object
Source	DataGen	JSON Object
Source	MQTT	JSON Object, Array of JSON
Source	PI Web (AF)	JSON Object
Source	PI Web (Tags)	JSON Object
Source	PI Web (Event Frames)	JSON Object
Destination	CONNECT Data Services	JSON Object, Array of JSON
Destination	CONNECT Data Services (Events)	JSON Object, Array of JSON
Destination	Azure Event Hub	JSON Object, Array of JSON
Destination	ESRI GeoEvent	JSON Object, Array of JSON
Destination	PI Web (Tags)	JSON Object, Array of JSON
Destination	pSQL	JSON Object, Array of JSON
Destination	MQTT	JSON Object, Array of JSON
Destination	Vantiq	JSON Object, Array of JSON

Note: Every CloudStream Destination is capable of processing single JSON Objects or an array of JSON Objects. CloudStream sources vary on the type of data they obtain, and MQTT data sources use the datatype of the data submitted by the user. However, invalid messages will be discarded by both Sources and Destinations.

Tab 2: Variables

The screenshot shows a window titled "Configure CloudStream Injector" with a close button (X) in the top right corner. It has two tabs: "Script" and "Variables", with "Variables" being the active tab. Below the tabs is a grey bar with the text "Initialize variables that will persist between each script invocation. (Optional)". Under this bar are two input fields: "Enter Variable Name" and "Enter Initial Value". Below these fields is a "Create Variable" button. Further down is a section titled "Current Variables" containing a table with three columns: "Name", "Initial Value", and "Delete". The table is currently empty, with the text "No data is available" centered below it. At the bottom of the window is a "Save and Exit" button.

Name	Initial Value	Delete
No data is available		

Users may define “Variables” whose values persist between each invocation of the injector script.

The first time each script is run (per service), the variable will contain the initial value set by the user. If the injector script modifies the variable’s value, then the new value will be stored in memory and available for the next time the script runs. A unique instance of each variable is saved per CloudStream service, e.g. if the same injector script is used in multiple services, one instance of the variable’s value will be held in memory per Service. Values are not shared or accessible across Services. Restarting the CloudStream application or deleting a service will reset the value of all Variables to their initial value.

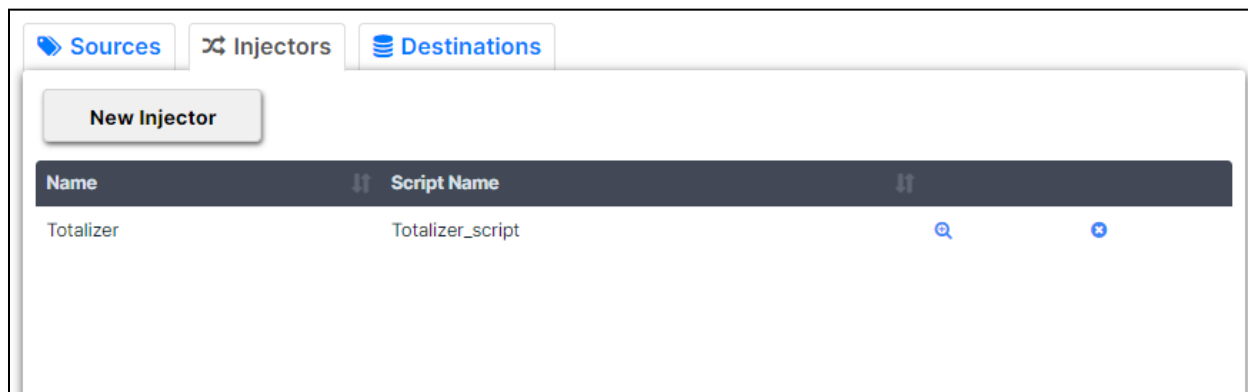
Note: As of CloudStream version 2.0, variables must be numeric or string datatypes. Additional datatypes will be supported in future versions of CloudStream. CloudStream will attempt to convert your variable’s initial value to a numeric, and on fail will treat the value as a string.

Instructions for this tab:

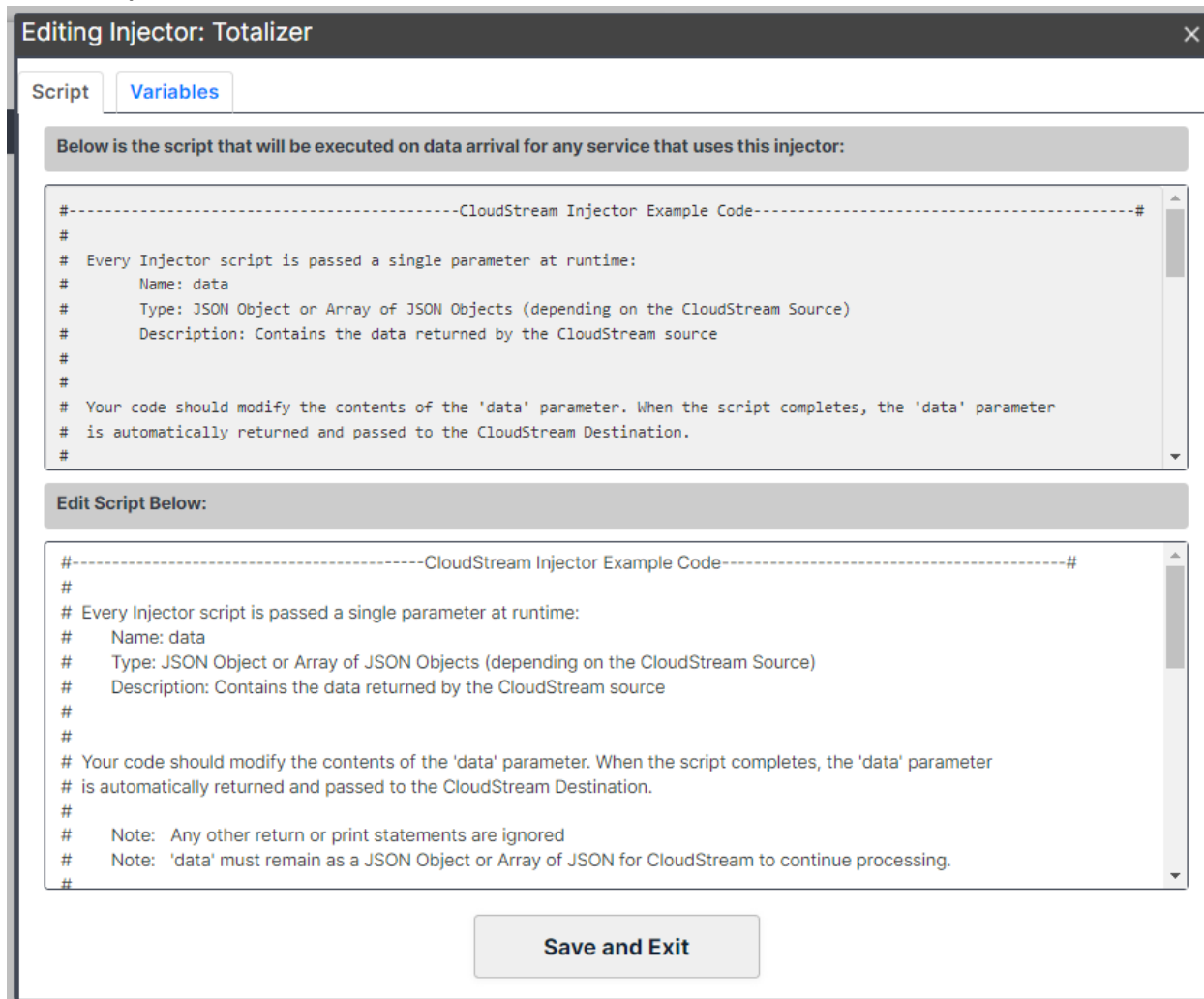
1. Enter the name for the Variable. This is how the Variable will be accessed from within the injector python script and retrieved/store between each invocation.
2. Enter the initial value for the Variable.
3. Click “Create Variable” to store the Variable within the Data Injector Config.
4. You may create multiple Variables, and you may delete Variables after they have been created.

5.2 Editing Data Injectors

Data Injector scripts may be edited at any time by pressing the **Magnifying Glass** button in the Injectors table:



Tab 1: Script



Instructions for this tab:

1. (Optional) Edit the python code for the selected Data Injector.
2. Press **Save and Exit** to save the changes and close the pop-up.
 - a. Script changes will take effect immediately in any active services.

Tab 2: Variables

Editing Injector: Totalizer

Script

Variables

Below are all the variables for this injector that will persist between script invocations.

Name	Initial Value
x	100
totalVal	0

Back

1

Next

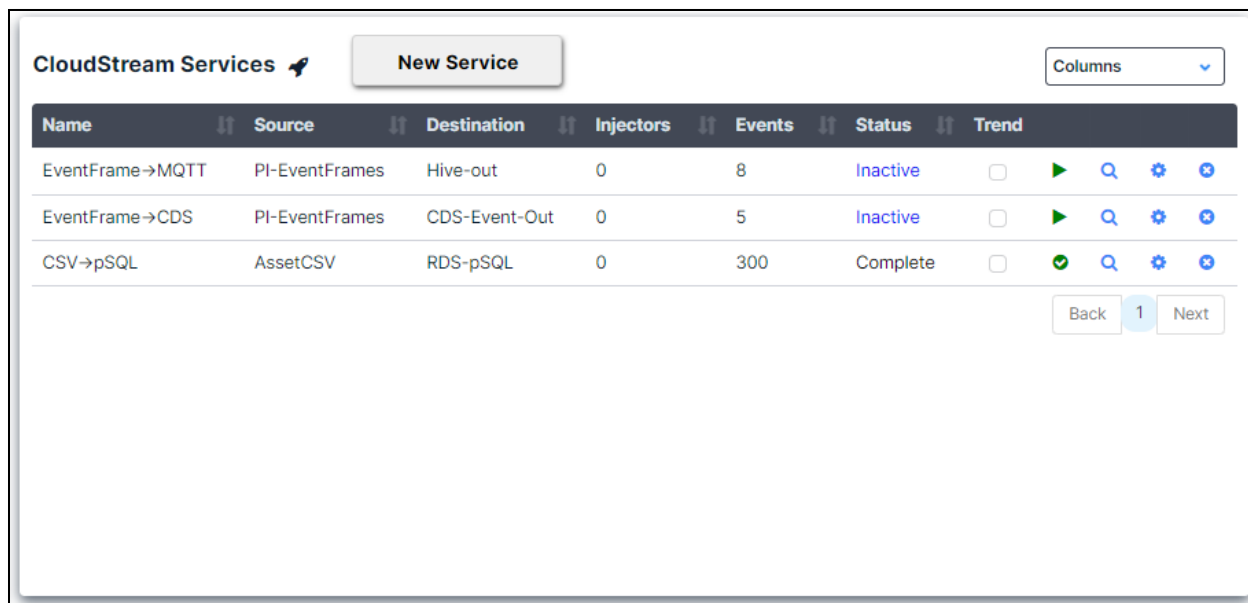
Instructions for this tab:

1. View any variables and their initial values configured for this script.

Note: As of CloudStream version 2.0, variables cannot be edited once created. Changing variables requires the Injector to be deleted and remade. Future versions of CloudStream will support ad-hoc variable updates.

6. CloudStream Service Creation

To create a CloudStream Service, click on the **“New Service”** button in the CloudStream Services table in the CloudStream System Control page:



The screenshot shows the 'CloudStream Services' page. At the top, there is a 'New Service' button. Below it is a table with the following columns: Name, Source, Destination, Injectors, Events, Status, and Trend. The table contains three rows of data.

Name	Source	Destination	Injectors	Events	Status	Trend
EventFrame→MQTT	PI-EventFrames	Hive-out	0	8	Inactive	☐
EventFrame→CDS	PI-EventFrames	CDS-Event-Out	0	5	Inactive	☐
CSV→pSQL	AssetCSV	RDS-pSQL	0	300	Complete	☒

At the bottom right of the table, there are buttons for 'Back', '1', and 'Next'.

This will open the Service Creation pop-up (next page)

Configure CloudStream Service

CloudStream Service Name:

Select Source:

--- No Source Selected ---

Select Injectors (Optional):

--- No Injector Selected ---

Order

Injector Name

Script Name

No data is available

Select Destination:

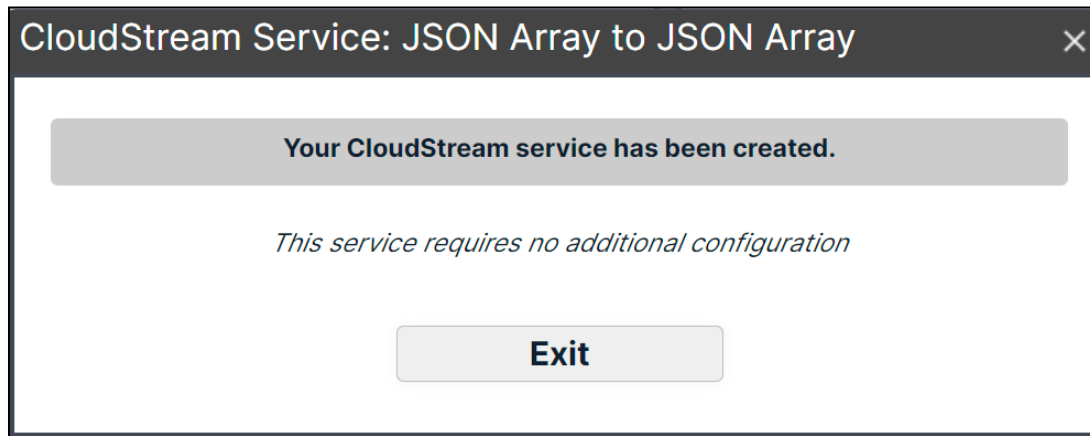
--- No Destination Selected ---

Create New Service

Instructions for creating a CloudStream Service:

1. Enter the name for the **CloudStream Service** into the top input box.
2. Select a **Source** from the dropdown.
3. Select a **Destination** from the dropdown.
4. **(Optional)** Sequentially add **Data Injectors** by selecting the desired injector from the dropdown and pressing the **Add** button. The order of the **Injectors** can be changed.
5. When complete, press **Create New Service**.

Upon successful completion, the following service pop-up will appear:



6.1 Service Monitoring

Live event counts for any service can be monitored by selecting the “Trend” option in the service table.

CloudStream Services 

New Service

Columns 

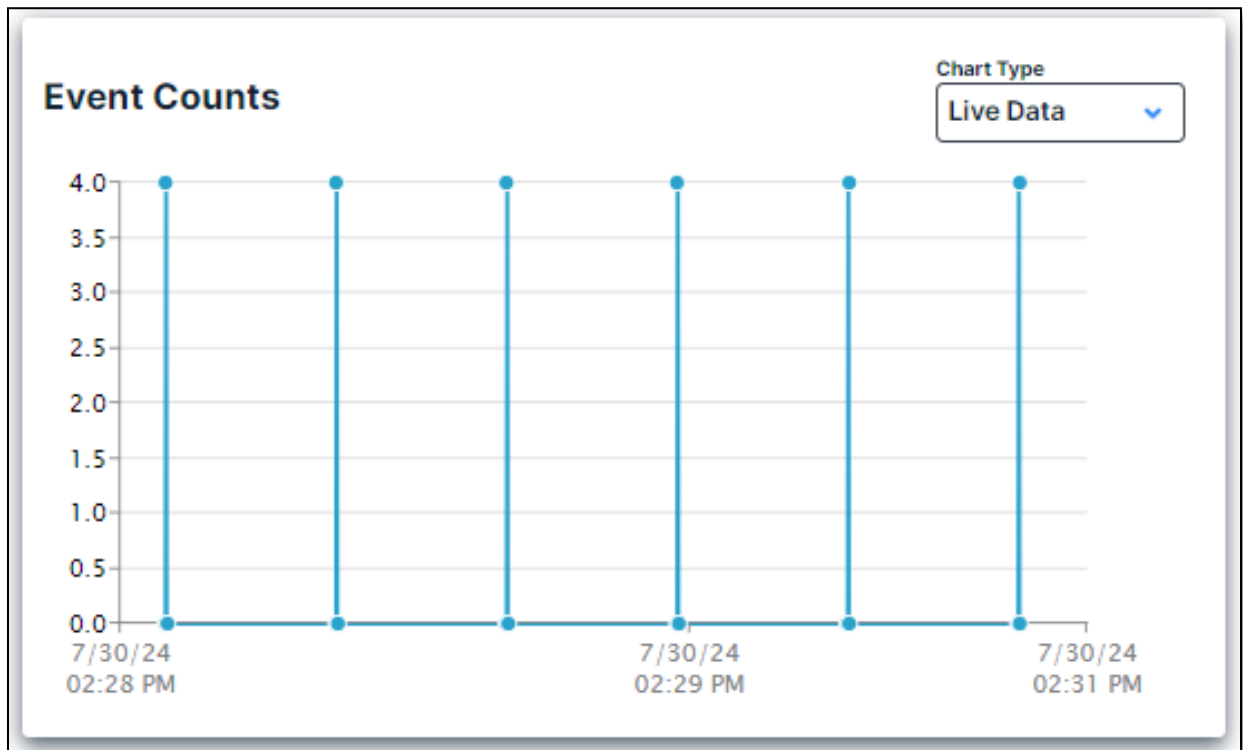
Name	Source	Destination	Injectors	Events	Status	Trend
EventFrame→MQTT	PI-EventFrames	Hive-out	0	8	Inactive	☐    
EventFrame→CDS	PI-EventFrames	CDS-Event-Out	0	5	Inactive	☐    
CSV→pSQL	AssetCSV	RDS-pSQL	0	300	Complete	☐    

Back

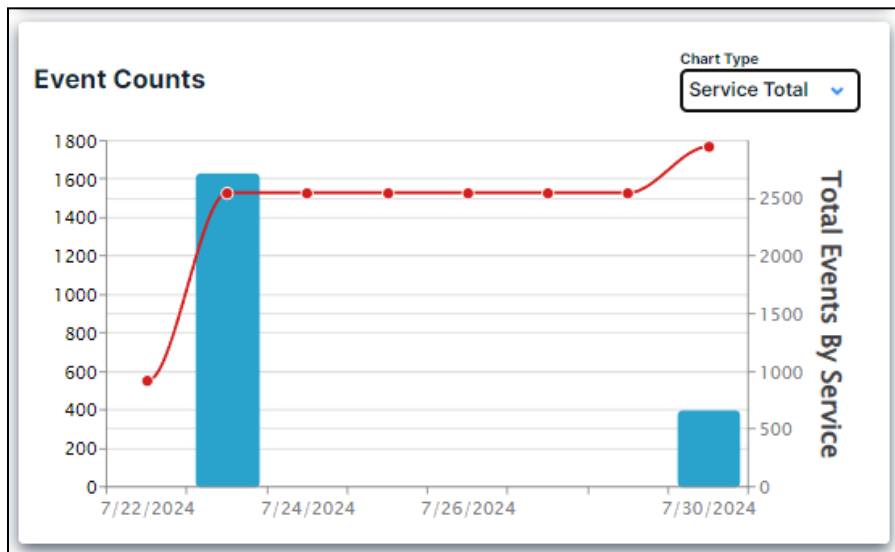
1

Next

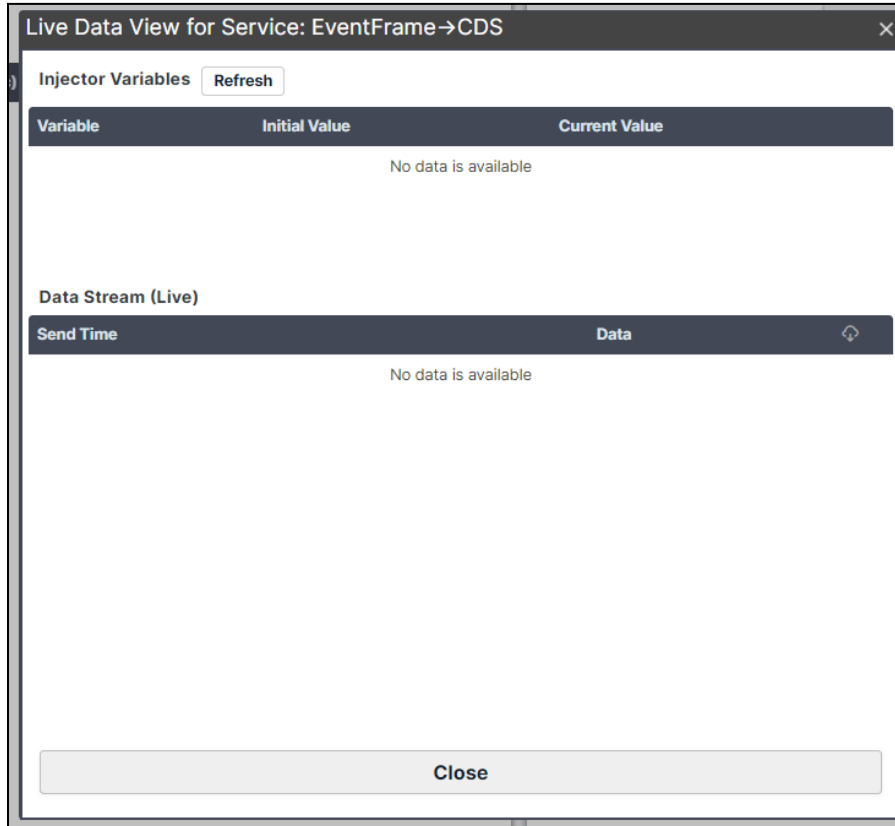
The Event Counts plot will automatically update whenever data is sent to the CloudStream destination within the selected service:



Changing the Chart Type from “Live Data” to “Service Total” will display the total data points sent per day and life total for the selected CloudStream Service. Up to 1 week of data is saved:



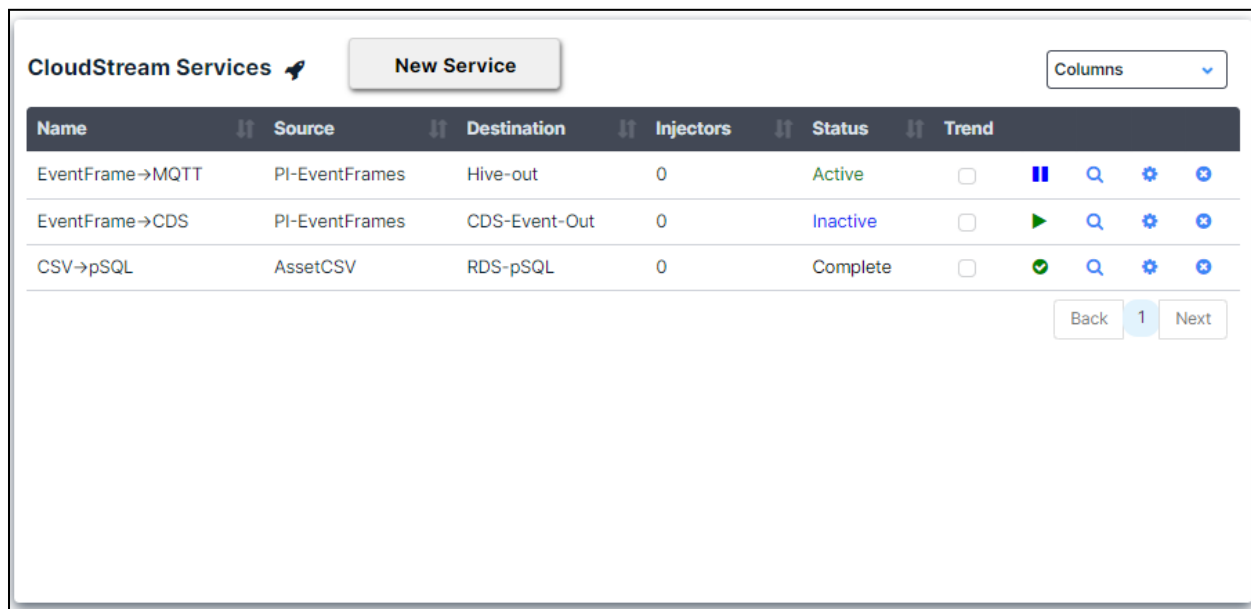
Clicking the magnifying glass icon in the service table will open the Live Data Viewer for the service. Data messages will be added to the screen in real-time as they are processed by Cloudstream. This pop-up also reports the current values for any Variables referenced by Data Injectors in the Service:



6.2 Service Control

Pause Service

CloudStream Services may be paused by clicking on the Pause icon in the service table:



Name	Source	Destination	Injectors	Status	Trend
EventFrame→MQTT	PI-EventFrames	Hive-out	0	Active	☐    
EventFrame→CDS	PI-EventFrames	CDS-Event-Out	0	Inactive	☐    
CSV→pSQL	AssetCSV	RDS-pSQL	0	Complete	☐    

While paused, services that poll data from Data Sources will stop polling. Services that receive data from remote sources will stop routing data.

Edit Service

CloudStream Services may be edited by clicking on the Gear button in the service table. Clicking this button will re-open the [service creation screen](#). Users may change Sources/Destinations, and may add, remove, or re-order Data Injectors.

Note: Editing a service will restart the state of the service, clearing any in-memory caches and may result in re-transmission of the most recent message.

Remove Service

CloudStream services may be deleted by clicking on the “X” button in the services table.

7. Revision History

Date	Author(s)	Changelist
7/30/2024	Renner Burkle Kai Chenhaun Varun Pardeshi	Initial Document for CloudStream v1.4
9/10/2024	Renner Burkle Kai Chenhaun Varun Pardeshi	V2.0 Release Updates